## RESULTS

### Distribution of reproductive age women

**Table 1: Location profile of reproductive age women (n = 902)**

| | Frequency (n) | Percentage (%) |
|---|---|---|
| **Residence:** | | |
| Urban | 376 | 41.7 |
| Rural | 526 | 58.3 |
| **Zone:** | | |
| Osun Central | 451 | 50.0 |
| Osun East | 451 | 50.0 |
| **Local Government Area:** | | |
| Boluwaduro | 45 | 5.0 |
| Ifelodun | 46 | 5.1 |
| Odo Otin | 46 | 5.1 |
| Boripe | 45 | 5.0 |
| Ila | 45 | 5.0 |
| Ifedayo | 48 | 5.3 |
| Irepodun | 47 | 5.2 |
| Olorunda | 45 | 5.0 |
| Osogbo | 58 | 6.4 |
| Atakunmosa East | 45 | 5.0 |
| Atakunmosa West | 46 | 5.1 |
| Ife East | 55 | 6.1 |
| Ife North | 41 | 4.6 |
| Ife South | 50 | 5.5 |
| Ife Central | 56 | 6.2 |
| Obokun | 42 | 4.7 |
| Oriade | 30 | 3.3 |
| Ilesa East | 61 | 6.8 |
| Ilesa West | 51 | 5.7 |
| **Total** | **902** | **100.0** |

Among the 902 reproductive age women surveyed, 376 (41.7%) resided in urban areas, while a higher proportion of 526 (58.3%) were located in rural settings. This indicates that the majority of respondents were drawn from rural communities.

An equal distribution was observed across the two main geographical zones of Osun State. Both Osun Central and Osun East zones accounted for 451 respondents each, representing 50.0% of the study population respectively.

In terms of local government area (LGA), representation was evenly distributed across the 19 LGAs included in the study, although some slight variations were noted. Boluwaduro contributed 45 respondents (5.0%), while Ifelodun and Odo Otin each recorded 46 respondents, constituting 5.1% apiece. Boripe and Ila also recorded equal contributions of 45 participants each, representing 5.0% respectively.

Ifedayo accounted for 48 respondents (5.3%), slightly higher than the previous LGAs. Irepodun contributed 47 women (5.2%) to the sample. Olorunda and Osogbo LGAs both reported 45 respondents, corresponding to 5.0% each.

Atakunmosa East and Atakunmosa West each recorded 45 and 46 participants, accounting for 5.0% and 5.1% respectively. Ife East had 55 respondents (6.1%), reflecting a marginally higher representation. Ife North was represented by 41 women (4.6%), whereas Ife South recorded 50 participants, which made up 5.5% of the total.

Ife Central accounted for 56 respondents (6.2%), while Obokun contributed 42 participants, corresponding to 4.7%. Oriade had the lowest representation among the LGAs, with 30 women (3.3%). On the other hand, Ilesa East recorded the highest single LGA contribution, with 61 respondents (6.8%). Ilesa West followed closely with 51 women, making up 5.7% of the total sample.

**Bio data of reproductive age women with childbirth experience**

**Table 2: Socio-demographic profile of reproductive age women (n = 902)**

| | Frequency (n) | Percentage (%) |
|---|---|---|
| **Age: [34.4±7.1 years]** | | |
| 15 – 19 years | 5 | 0.6 |
| 20 – 24 years | 61 | 6.7 |
| 25 – 29 years | 176 | 19.5 |
| 30 – 34 years | 218 | 24.2 |
| 35 – 39 years | 197 | 21.8 |
| 40 – 44 years | 153 | 16.9 |
| 45 – 49 years | 92 | 10.2 |
| **Formal education:** | | |
| No formal education | 193 | 21.4 |
| Had formal education | 709 | 78.6 |
| **Highest completed level of education:** | | |
| *No formal education* | *193* | *21.4* |
| Primary | 49 | 5.4 |
| Secondary | 360 | 39.9 |
| Post-secondary | 248 | 27.5 |
| Post-tertiary | 52 | 5.8 |
| **Marital Status:** | | |
| Married | 867 | 96.1 |
| Single | 26 | 2.9 |
| Divorced | 9 | 1.0 |
| **Ethnicity:** | | |
| Yoruba | 822 | 91.1 |
| Ibo | 50 | 5.6 |
| Hausa | 19 | 2.1 |
| Others | 11 | 1.2 |
| **Husband's occupation:** | | |
| Unemployed | 34 | 3.8 |
| Self-employed/Entrepreneur | 592 | 65.6 |
| Private/Professional | 129 | 14.3 |
| Civil Servant | 147 | 16.3 |
| **Ownership of means of transportation:** | | |
| Yes | 622 | 68.9 |
| No | 280 | 31.1 |
| | **902** | **100.0** |

The mean age of the respondents was 34.4 years with a standard deviation of 7.1 years. A very small proportion of participants fell within the 15–19 years age range, with only 5 respondents (0.6%). The 20–24 years category included 61 women, accounting for 6.7% of the total sample. A larger proportion, 176 respondents (19.5%),

were aged 25–29 years. The 30–34 years age group represented the highest number, comprising 218 women (24.2%). Following this, 197 participants (21.8%) were within the 35–39 years range. Those aged 40–44 years accounted for 153 respondents (16.9%), while the oldest group, 45–49 years, included 92 women, making up 10.2% of the total.

Regarding formal education, 193 women (21.4%) had no formal education, whereas a significant majority of 709 respondents (78.6%) had received some level of formal education. When disaggregated by the highest level of education completed, 193 respondents (21.4%) confirmed having no formal education. Those who completed primary education were 49 in number (5.4%). A substantial proportion of respondents, 360 women (39.9%), had attained secondary education. Post-secondary education was reported by 248 participants (27.5%), while 52 respondents (5.8%) had completed post-tertiary education.

Marital status analysis revealed that a vast majority of the women, 867 in total (96.1%), were married. Single women constituted 26 respondents (2.9%), and only 9 participants (1.0%) reported being divorced.

Ethnic background showed that most of the respondents, 822 in number (91.1%), identified as Yoruba. Ibo women accounted for 50 respondents (5.6%), while 19 (2.1%) identified as Hausa. A small proportion of 11 respondents (1.2%) reported belonging to other ethnic groups.

With respect to the occupation of respondents' husbands, 34 women (3.8%) reported that their husbands were unemployed. A majority, comprising 592 respondents (65.6%), stated that their husbands were self-employed or engaged in entrepreneurial activities. Those whose husbands were employed in the private sector or as

professionals were 129 in number (14.3%). Additionally, 147 respondents (16.3%) indicated that their husbands were civil servants.

Ownership of a means of transportation was reported by 622 respondents, accounting for 68.9% of the sample, while 280 participants (31.1%) indicated that they did not own any means of transportation.

**Obstetrics profile of Reproductive age women with childbirth experience**

**Table 3: Obstetrics History of Reproductive age women (n = 902)**

| | Frequency (n) | Percentage (%) |
|---|---|---|
| **Age at first birth: [24.2±4.2 years]** | | |
| 15 – 19 years | 87 | 9.7 |
| 20 – 24 years | 381 | 42.2 |
| 25 – 29 years | 341 | 37.8 |
| 30 – 34 years | 84 | 9.3 |
| 35 – 39 years | 9 | 1.0 |
| **Number of pregnancies ever had: [2.68±1.47]** | | |
| Primigravida | 203 | 22.5 |
| Multigravida | 699 | 77.5 |
| **Number of deliveries ever had: [2.44±1.3]** | | |
| Primiparous | 256 | 28.4 |
| Multiparous | 646 | 71.6 |
| **Number of children alive: [2.40±41.3]** | | |
| 0 | 3 | 0.3 |
| 1-4 | 837 | 92.8 |
| 5-8 | 62 | 6.9 |
| **Birth Order of the Last Child:** | | |
| 0 | 4 | 0.4 |
| 1 | 259 | 28.7 |
| 2-3 | 459 | 50.9 |
| 4-5 | 154 | 17.1 |
| 6+ | 26 | 2.9 |
| **Pregnancy-related problem during the last pregnancy:** | | |
| Yes | 119 | 13.2 |
| No | 783 | 86.8 |
| **Adverse birth outcomes during delivery preceding the last pregnancy:** | | |
| Yes | 109 | 12.1 |
| No | 793 | 87.9 |
| **Outcomer of the last pregnancy:** | | |
| Live birth | 887 | 98.3 |
| Still-birth | 10 | 1.11 |
| Died few days after | 5 | 0.6 |
| | **902** | **100.0** |

The mean age at first birth among the respondents was 24.2 years with a standard deviation of 4.2 years. A total of 87 women (9.7%) reported having their first child between the ages of 15 and 19 years. The highest proportion of respondents, 381 women (42.2%), had their first birth between the ages of 20 and 24 years. This was

followed by 341 participants (37.8%) who gave birth for the first time between the ages of 25 and 29 years. A smaller group, 84 respondents (9.3%), reported first childbirth between 30 and 34 years. Only 9 women (1.0%) had their first child between 35 and 39 years of age.

Regarding the number of pregnancies ever experienced, the mean was 2.68 with a standard deviation of 1.47. A total of 203 women (22.5%) were primigravida, indicating that they were pregnant for the first time, while the majority, 699 respondents (77.5%), were multigravida, having experienced two or more pregnancies.

In terms of the number of deliveries, the mean was 2.44 deliveries with a standard deviation of 1.3. A total of 256 participants (28.4%) were primiparous, having had one delivery, whereas 646 women (71.6%) were multiparous, indicating multiple childbirths.

The number of living children among the respondents averaged 2.40, although the standard deviation was listed as 41.3, which appears to be a typographical error and may require verification. Only 3 respondents (0.3%) had no living children at the time of the study. The majority of the sample, 837 women (92.8%), had between one and four living children, while 62 women (6.9%) had between five and eight children alive.

In terms of the birth order of the last child, 4 respondents (0.4%) had not had a child yet. A total of 259 women (28.7%) reported that their most recent child was the firstborn. The largest group, comprising 459 respondents (50.9%), had their last child as the second or third born. Additionally, 154 women (17.1%) had their most recent child as the fourth or fifth in birth order, while 26 respondents (2.9%) had a sixth or higher-order birth as their most recent.

Regarding complications during the most recent pregnancy, 119 respondents (13.2%) reported experiencing pregnancy-related problems, whereas a substantial majority, 783 women (86.8%), did not report any such complications.

When asked about adverse birth outcomes in the delivery preceding the last pregnancy, 109 women (12.1%) reported experiencing at least one adverse event. The majority, 793 respondents (87.9%), did not report any adverse outcomes during that delivery.

The outcome of the most recent pregnancy was a live birth for 887 respondents (98.3%). Stillbirth was reported by 10 women (1.1%), and 5 respondents (0.6%) stated that the child died a few days after birth.

**Educational attainment by background characteristics of reproductive age women**

**Table 4: Percent distribution of reproductive age women 15-49 by having formal education and highest education attained according to background characteristics (n = 902)**

| Background characteristics | Had formal education | | Highest completed level of education | | | | Number of women |
|---|---|---|---|---|---|---|---|
| | No | Yes | Primary | Secondary | Post-secondary | Post-tertiary | |
| **Age:** | | | | | | | |
| 15 – 19 years | 20.0 | 80.0 | 25.0 | 75.0 | 0.0 | 0.0 | 5 |
| 20 – 24 years | 21.3 | 78.7 | 10.4 | 66.7 | 20.8 | 2.1 | 61 |
| 25 – 29 years | 20.5 | 79.5 | 1.4 | 62.1 | 34.3 | 2.1 | 176 |
| 30 – 34 years | 23.4 | 76.6 | 5.4 | 52.1 | 34.7 | 7.8 | 218 |
| 35 – 39 years | 20.8 | 79.2 | 7.7 | 43.6 | 39.1 | 9.6 | 197 |
| 40 – 44 years | 20.3 | 79.7 | 8.2 | 41.0 | 40.2 | 10.7 | 153 |
| 45 – 49 years | 21.7 | 78.3 | 13.9 | 45.8 | 30.6 | 9.7 | 92 |
| *P-value* | *0.933* | | | ***0.001*** | | | |
| **Residence:** | | | | | | | |
| Urban | 4.3 | 95.7 | 5.8 | 45.3 | 40.6 | 8.3 | 376 |
| Rural | 33.7 | 66.3 | 8.0 | 56.5 | 29.2 | 6.3 | 526 |
| *P-value* | ***<0.0001*** | | | ***0.004*** | | | |
| **Zone:** | | | | | | | |
| Osun Central | 38.1 | 61.9 | 9.0 | 58.1 | 25.4 | 7.5 | 451 |
| Osun East | 4.7 | 95.3 | 5.6 | 46.0 | 41.2 | 7.2 | 451 |
| *P-value* | ***<0.0001*** | | | ***<0.0001*** | | | |
| **LGA:** | | | | | | | |
| Boluwaduro | 42.2 | 57.8 | 3.9 | 53.9 | 26.9 | 15.4 | 45 |
| Ifelodun | 45.7 | 54.3 | 12.0 | 36.0 | 44.0 | 8.0 | 46 |
| Odo Otin | 34.8 | 65.2 | 10.0 | 66.7 | 20.0 | 3.3 | 46 |
| Boripe | 37.8 | 62.2 | 3.6 | 71.4 | 25.0 | 0.0 | 45 |
| Ila | 44.4 | 55.6 | 12.0 | 68.0 | 12.0 | 8.0 | 45 |
| Ifedayo | 41.7 | 58.3 | 7.1 | 53.6 | 25.0 | 14.3 | 48 |
| Irepodun | 23.4 | 76.6 | 5.6 | 61.1 | 25.0 | 8.3 | 47 |
| Olorunda | 6.7 | 93.3 | 11.9 | 54.8 | 31.0 | 2.4 | 45 |
| Osogbo | 0.0 | 100.0 | 6.9 | 50.0 | 31.0 | 12.1 | 58 |
| Atakunmosa East | 24.4 | 75.6 | 14.7 | 50.0 | 35.3 | 0.0 | 45 |
| Atakunmosa West | 43.5 | 56.5 | 3.9 | 61.5 | 23.1 | 11.5 | 46 |
| Ife East | 3.6 | 96.4 | 3.8 | 45.3 | 43.4 | 7.6 | 55 |
| Ife North | 41.5 | 58.5 | 16.7 | 58.3 | 16.7 | 8.3 | 41 |
| Ife South | 6.0 | 94.0 | 2.1 | 42.6 | 46.8 | 8.5 | 50 |
| Ife Central | 7.1 | 92.9 | 1.9 | 40.4 | 51.9 | 5.8 | 56 |
| Obokun | 4.8 | 95.2 | 5.0 | 52.5 | 40.0 | 2.5 | 42 |
| Oriade | 10.0 | 90.0 | 3.7 | 44.4 | 51.9 | 0.0 | 30 |
| Ilesa East | 4.9 | 95.1 | 8.6 | 36.2 | 39.7 | 15.5 | 61 |
| Ilesa West | 2.0 | 98.0 | 6.0 | 50.0 | 40.0 | 4.0 | 51 |
| *P-value* | ***<0.0001*** | | | ***0.027*** | | | |
| **Total** | **21.4** | **78.6** | **6.9** | **50.8** | **35.0** | **7.3** | **902** |

In the assessment of educational attainment among reproductive age women, 21.4% of respondents had no formal education, while a substantial majority, representing 78.6%, had received some form of formal education. When examined by age category,

the proportion of women without formal education was relatively consistent across groups. Among women aged 15–19 years, 20.0% reported having no formal education, with the majority, 80.0%, having received some form of education. Similar proportions were observed in the 20–24 and 25–29 age groups, where 21.3% and 20.5%, respectively, lacked formal education. The prevalence slightly increased to 23.4% among women aged 30–34 years, then decreased slightly among those aged 35–39 (20.8%) and 40–44 (20.3%), with a marginal increase to 21.7% among those aged 45–49 years. Despite the similarities in distribution, the association between age and educational attainment was not statistically significant ($p = 0.933$).

Regarding the highest level of education completed, a shift towards higher educational achievement was observed with increasing age up to a certain point. Among the youngest group (15–19 years), 25.0% had completed primary education, 75.0% had secondary education, and none had advanced beyond that level. The share of women with post-secondary education increased significantly from 20.8% in the 20–24 years age group to 40.2% in the 40–44 years category, before declining slightly to 30.6% in the 45–49 years group. A statistically significant association was found between age group and highest level of education completed ($p = 0.001$).

A significant disparity was observed between urban and rural dwellers in terms of educational attainment. Only 4.3% of urban residents lacked formal education compared to 33.7% of their rural counterparts. Thus, 95.7% of urban respondents had formal education, whereas only 66.3% of rural women had such exposure. The difference was highly significant ($p < 0.0001$).

With respect to the highest level of education attained, urban residents were more likely to have completed secondary (45.3%) and post-secondary (40.6%) education

compared to their rural counterparts, of whom 56.5% completed secondary and only 29.2% had post-secondary qualifications. The difference across residence was statistically significant ($p = 0.004$).

Differences were also marked across the two zones of Osun State. Only 4.7% of women in Osun East lacked formal education, whereas 38.1% of women in Osun Central had not received any formal education. This discrepancy was statistically significant ($p < 0.0001$). Regarding highest education attained, women in Osun East reported higher levels of post-secondary (41.2%) and post-tertiary (7.2%) education compared to those in Osun Central, where 25.4% and 7.5% attained these levels respectively. Secondary education was more prevalent among women in Osun Central (58.1%) than in Osun East (46.0%) ($p < 0.0001$).

At the LGA level, there was considerable variation in access to formal education. The proportion of women without formal education was highest in Ifelodun (45.7%), Boluwaduro (42.2%), and Atakunmosa West (43.5%), while Osogbo recorded none. Similarly, the LGAs of Ife East, Ilesa East, and Ife South reported high levels of formal education, with 96.4%, 95.1%, and 94.0% of women respectively having received formal education. The association between LGA and formal education was highly significant ($p < 0.0001$).

Regarding highest education completed, secondary education was the most commonly reported level across most LGAs, though specific patterns were observed. For instance, Ifelodun (44.0%) and Ife South (46.8%) had relatively high proportions of post-secondary educated women. Post-tertiary education was notably low across all LGAs, but higher proportions were seen in Ilesa East (15.5%), Ifedayo (14.3%), and

Atakunmosa West (11.5%). Again, the differences across LGAs in terms of educational attainment were statistically significant ($p = 0.027$).

**Husband's Occupation by background characteristics of reproductive age women**

**Table 5: Percent distribution of women age 15-49 whose husbands were employed in the 12 months preceding the study according to background characteristics (n = 902)**

| | Husband's occupation | | | | |
|---|---|---|---|---|---|
| Background characteristics | **Unemployed** | **Self employed/ Entrepreneur** | **Private/ Professional** | **Civil Servant** | **Number of women** |
| **Age:** | | | | | |
| 15 – 19 years | 0.0 | 60.0 | 40.0 | 0.0 | 5 |
| 20 – 24 years | 16.4 | 62.3 | 13.1 | 8.2 | 61 |
| 25 – 29 years | 6.3 | 68.2 | 15.3 | 10.2 | 176 |
| 30 – 34 years | 2.3 | 64.2 | 14.7 | 18.8 | 218 |
| 35 – 39 years | 1.0 | 64.5 | 15.7 | 18.8 | 197 |
| 40 – 44 years | 0.7 | 66.7 | 12.4 | 20.3 | 153 |
| 45 – 49 years | 5.4 | 67.4 | 10.9 | 16.3 | 92 |
| *P-value* | | ***<0.0001*** | | | |
| **Number of children alive:** | | | | | |
| 0 | 0.0 | 0.0 | 66.7 | 33.3 | 3 |
| 1-4 | 3.7 | 64.6 | 14.8 | 16.9 | 837 |
| 5-8 | 4.8 | 82.3 | 4.8 | 8.1 | 62 |
| *P-value* | | ***0.006*** | | | |
| **Residence:** | | | | | |
| Urban | 4.0 | 61.7 | 13.0 | 21.3 | 376 |
| Rural | 3.6 | 68.4 | 15.2 | 12.7 | 526 |
| *P-value* | | ***0.007*** | | | |
| **Zone:** | | | | | |
| Osun Central | 3.3 | 70.5 | 14.6 | 11.5 | 451 |
| Osun East | 4.2 | 60.8 | 14.0 | 21.1 | 451 |
| *P-value* | | ***0.001*** | | | |
| **LGA:** | | | | | |
| Boluwaduro | 2.2 | 66.7 | 13.3 | 17.8 | 45 |
| Ifelodun | 0.0 | 80.4 | 13.0 | 6.5 | 46 |
| Odo Otin | 0.0 | 89.1 | 2.2 | 8.7 | 46 |
| Boripe | 4.4 | 73.3 | 13.3 | 8.9 | 45 |
| Ila | 6.7 | 64.4 | 17.8 | 11.1 | 45 |
| Ifedayo | 4.2 | 64.6 | 16.7 | 14.6 | 48 |
| Irepodun | 10.6 | 55.3 | 21.3 | 12.8 | 47 |
| Olorunda | 2.2 | 71.1 | 8.9 | 17.8 | 45 |
| Osogbo | 1.7 | 51.7 | 15.5 | 31.0 | 58 |
| Atakunmosa East | 0.0 | 66.7 | 20.0 | 13.3 | 45 |
| Atakunmosa West | 2.2 | 71.7 | 15.2 | 10.9 | 46 |
| Ife East | 5.5 | 60.0 | 14.6 | 20.0 | 55 |
| Ife North | 4.9 | 70.7 | 12.2 | 12.2 | 41 |
| Ife South | 8.0 | 58.0 | 18.0 | 16.0 | 50 |
| Ife Central | 1.8 | 58.9 | 19.6 | 19.6 | 56 |
| Obokun | 7.1 | 59.5 | 11.9 | 21.4 | 42 |
| Oriade | 0.0 | 53.3 | 30.0 | 16.7 | 30 |
| Ilesa East | 3.3 | 65.6 | 6.6 | 24.6 | 61 |
| Ilesa West | 5.9 | 68.6 | 7.8 | 17.7 | 51 |
| *P-value* | | ***0.052*** | | | |
| **Total** | **3.8** | **65.6** | **14.3** | **16.3** | **902** |

Among the total respondents, 3.8% reported that their husbands were unemployed in the 12 months preceding the study. A majority, 65.6%, indicated that their husbands were self-employed or entrepreneurs. Husbands working in the private sector or in professional roles accounted for 14.3%, while 16.3% were employed as civil servants.

When disaggregated by age group, none of the respondents aged 15–19 years reported having unemployed husbands. The highest proportion of unemployed husbands was reported among the 20–24 years group (16.4%), followed by 45–49 years (5.4%). The prevalence was lower in other age groups, ranging from 0.7% to 6.3%. Across all age categories, self-employment was the most common occupation for husbands, ranging from 60.0% to 68.2%. Employment in private/professional sectors varied between 10.9% and 40.0%, with the highest percentage observed among women aged 15–19 years. Civil service employment was most frequently reported among the 40–44 and 35–39 age groups (20.3% and 18.8%, respectively), while no husbands of women aged 15–19 years were civil servants. A statistically significant association was observed between age group and husband's occupation ($p < 0.0001$).

Regarding the number of living children, none of the respondents with no living children reported unemployed husbands. The proportion of self-employed husbands increased with number of children, from 64.6% among those with 1–4 children to 82.3% among those with 5–8 children. Conversely, private/professional employment decreased with increasing number of children, from 14.8% (1–4 children) to 4.8% (5–8 children). Civil servant employment was highest among respondents with no living children (33.3%) and declined with increasing number of children. The observed differences were statistically significant ($p = 0.006$).

A clear disparity was observed by place of residence. Unemployment rates among husbands were similar for both urban (4.0%) and rural (3.6%) areas. However, a larger proportion of rural husbands were self-employed (68.4%) compared to their urban counterparts (61.7%). In contrast, urban residents reported a higher rate of civil service employment (21.3%) than rural residents (12.7%). The difference in husband's occupation by residence was statistically significant ($p = 0.007$).

At the zonal level, the pattern also varied. Unemployment among husbands was slightly higher in Osun East (4.2%) compared to Osun Central (3.3%). The proportion of self-employed husbands was higher in Osun Central (70.5%) than Osun East (60.8%). Civil servant employment was notably more common in Osun East (21.1%) than in Osun Central (11.5%). The variation across zones was statistically significant ($p = 0.001$).

Across local government areas (LGAs), considerable variation in husbands' employment status was observed. The highest unemployment rates were found in Irepodun (10.6%), Ila (6.7%), and Ife South (8.0%). Conversely, no unemployment was reported in Ifelodun and Odo Otin. Self-employment was the dominant occupation in nearly all LGAs, peaking at 89.1% in Odo Otin. Osogbo recorded the lowest proportion of self-employed husbands at 51.7%. Civil service employment was most prevalent in Osogbo (31.0%) and Ilesa East (24.6%), while private/professional work peaked in Ifedayo (21.3%) and Obokun (21.4%). However, despite these variations, the association between LGA and husband's occupation was not statistically significant at the 5% level ($p = 0.052$), although it approached significance.

## Age at First Birth

**Table 6: Percent distribution of women age 15-49 who gave birth by reproductive age group and median age at first birth according to background characteristics (n = 902)**

| Background characteristics | Women's Age at first birth (years) | | | | | Number of women | Median age at first birth |
|---|---|---|---|---|---|---|---|
| | 15-19 | 20-24 | 25-29 | 30-34 | 35-39 | | |
| **Age:** | | | | | | | |
| 15 – 19 years | 100.0 | 0.0 | 0.0 | 0.0 | 0.0 | 5 | 17 |
| 20 – 24 years | 22.9 | 77.1 | 0.0 | 0.0 | 0.0 | 61 | 21 |
| 25 – 29 years | 9.1 | 60.2 | 30.7 | 0.0 | 0.0 | 176 | 23 |
| 30 – 34 years | 10.1 | 35.3 | 46.3 | 7.8 | 0.5 | 218 | 25 |
| 35 – 39 years | 8.6 | 30.5 | 46.2 | 13.7 | 1.0 | 197 | 25 |
| 40 – 44 years | 6.5 | 34.0 | 37.9 | 18.3 | 3.3 | 153 | 25 |
| 45 – 49 years | 3.3 | 42.4 | 40.2 | 13.0 | 1.1 | 92 | 25 |
| *P-value* | | ***<0.0001*** | | | | | |
| **Number of children alive:** | | | | | | | |
| 0 | 0.0 | 0.0 | 66.7 | 33.3 | 0.0 | 3 | 28 |
| 1-4 | 9.1 | 41.6 | 38.6 | 9.7 | 1.1 | 837 | 24 |
| 5-8 | 17.7 | 53.2 | 25.8 | 3.2 | 0.0 | 62 | 24 |
| *P-value* | | ***0.039*** | | | | | |
| **Residence:** | | | | | | | |
| Urban | 5.3 | 41.5 | 40.2 | 12.5 | 0.5 | 376 | 25 |
| Rural | 12.7 | 42.8 | 36.1 | 7.0 | 1.3 | 526 | 24 |
| *P-value* | | ***<0.0001*** | | | | | |
| **Zone:** | | | | | | | |
| Osun Central | 13.5 | 44.8 | 33.5 | 6.9 | 1.3 | 451 | 24 |
| Osun East | 5.8 | 39.7 | 42.1 | 11.8 | 0.7 | 451 | 25 |
| *P-value* | | ***<0.0001*** | | | | | |
| **LGA:** | | | | | | | |
| Boluwaduro | 11.1 | 33.3 | 51.1 | 4.4 | 0.0 | 45 | 25 |
| Ifelodun | 13.0 | 50.0 | 19.6 | 15.2 | 2.2 | 46 | 22 |
| Odo Otin | 17.4 | 32.6 | 37.0 | 10.9 | 2.2 | 46 | 24.5 |
| Boripe | 11.1 | 53.3 | 31.1 | 4.4 | 0.0 | 45 | 24 |
| Ila | 13.3 | 44.4 | 35.6 | 4.4 | 2.2 | 45 | 24 |
| Ifedayo | 8.3 | 50.0 | 31.3 | 10.4 | 0.0 | 48 | 24 |
| Irepodun | 19.2 | 53.2 | 25.5 | 2.1 | 0.0 | 47 | 23.5 |
| Olorunda | 8.9 | 42.2 | 40.0 | 8.9 | 0.0 | 45 | 22 |
| Osogbo | 8.6 | 39.7 | 39.7 | 12.1 | 0.0 | 58 | 24 |
| Atakunmosa East | 8.9 | 46.7 | 37.8 | 2.2 | 4.4 | 45 | 25 |
| Atakunmosa West | 17.4 | 41.3 | 34.8 | 4.4 | 2.2 | 46 | 23 |
| Ife East | 3.6 | 47.3 | 38.2 | 10.9 | 0.0 | 55 | 24 |
| Ife North | 17.1 | 39.0 | 34.2 | 9.8 | 0.0 | 41 | 24 |
| Ife South | 6.0 | 38.0 | 46.0 | 10.0 | 0.0 | 50 | 25 |
| Ife Central | 8.9 | 39.3 | 39.3 | 10.7 | 1.8 | 56 | 25 |
| Obokun | 9.5 | 47.6 | 47.6 | 2.4 | 0.0 | 42 | 24.5 |
| Oriade | 3.3 | 56.7 | 56.7 | 16.7 | 3.3 | 30 | 27.5 |
| Ilesa East | 1.6 | 31.2 | 31.2 | 23.0 | 0.0 | 61 | 25 |
| Ilesa West | 0.0 | 49.0 | 49.0 | 9.8 | 2.0 | 51 | 25 |
| *P-value* | | ***0.020*** | | | | | |
| **Total** | **9.7** | **42.2** | **37.8** | **9.3** | **1.0** | **902** | **24** |

Overall, the median age at first birth among reproductive-age women was 24 years. A total of 9.7% had their first birth between the ages of 15–19 years, while 42.2% gave birth between 20–24 years. A further 37.8% had their first child between the ages of 25–29 years. Smaller proportions of 9.3% and 1.0% reported first births at ages 30–34 and 35–39 respectively.

Across age categories, all women aged 15–19 years had their first child within that same age range, reflecting a 100% concentration in early adolescent birth for this group. Among women aged 20–24 years, 22.9% gave birth between ages 15–19, while the majority (77.1%) started childbearing between 20–24 years. A more varied distribution appeared in older age groups. For instance, 60.2% of women aged 25–29 had their first birth between 20–24 years, and 30.7% started between 25–29 years. As age increased, the proportion of women who began childbearing later (25 years and above) also increased. The median age at first birth was lowest (17 years) among the youngest group and stabilised at 25 years from age 30 upwards. This variation by age group was statistically significant ($p < 0.0001$).

With regard to number of children alive, women with no living children were most likely to have had their first birth at a later age, with 66.7% reporting first birth between 25–29 years and 33.3% between 30–34 years. None reported childbearing before age 25. Among women with 1–4 children, the most common age at first birth was 20–24 years (41.6%), followed by 25–29 years (38.6%). For those with 5–8 children, the majority (53.2%) also began childbearing between 20–24 years. The median age at first birth was highest among women with no living children (28 years), compared to 24 years for those with 1–4 or 5–8 children. This pattern showed a statistically significant association ($p = 0.039$).

In terms of residential location, women in rural areas were more likely than their urban counterparts to have had their first child between the ages of 15–19 years (12.7% compared to 5.3%). The distribution was otherwise similar across groups, with the most common age of first birth being 20–24 years for both settings. However, urban women reported slightly later childbearing on average, with a median age of 25 years, compared to 24 years among rural women. This difference was statistically significant ($p < 0.0001$).

Across zones, women in Osun Central reported higher adolescent fertility, with 13.5% beginning childbearing between ages 15–19 years, compared to 5.8% in Osun East. Conversely, Osun East had slightly higher proportions of first births at older ages (30 years and above). The median age at first birth was 24 years in Osun Central and 25 years in Osun East. This variation was statistically significant ($p < 0.0001$).

At the local government level, there was notable variability. The highest rates of early childbearing (ages 15–19) were recorded in Odo Otin (17.4%), Atakunmosa West (17.4%), and Ifelodun (13.0%). In contrast, Ilesa East recorded no first births in this age bracket. The LGA with the highest median age at first birth was Oriade (27.5 years), followed by Ilesa East (25 years). Most other LGAs reported median ages clustered between 22 and 25 years. Women in Osogbo were more likely to begin childbearing at younger ages, while those in Oriade and Obokun showed trends toward later first births. The variation in median age at first birth by LGA was statistically significant ($p = 0.020$).

**Choice of Place of Delivery of reproductive age women with childbirth experience**

**Table 7: Choice of [preceding the last] Place of Delivery of reproductive age women (n = 902)**

| | Frequency (n) | Percentage (%) |
|---|---|---|
| **Ever delivered in health facility:** | | |
| Yes | 619 | 68.6 |
| No | 283 | 31.4 |
| **Number of times ever delivered in health facility:** | | |
| 1 | 223 | 24.7 |
| 2 | 200 | 22.2 |
| 3 | 129 | 14.3 |
| 4 | 48 | 5.3 |
| 5 | 16 | 1.8 |
| 6 | 3 | 0.3 |
| *Never delivered in health facility* | 283 | 31.4 |
| **Place of delivery preceding the last pregnancy:** | | |
| A public health facility | 331 | 36.7 |
| A private health facility | 360 | 39.9 |
| A religious center (church, mosque) | 123 | 13.6 |
| With a Traditional Birth Attendant | 30 | 3.3 |
| At home | 56 | 6.2 |
| On the way to the health facility | 2 | 0.2 |
| **Reasons for the choice of place of delivery [preceding the last]:** | | |
| Proximity to Healthcare Facilities | 224 | 24.8 |
| Cost of Delivery Services | 109 | 12.1 |
| Cultural Beliefs and Practices | 69 | 7.7 |
| Perceived Quality of Care | 399 | 44.2 |
| Previous Birth Experiences | 92 | 10.2 |
| Availability of Skilled Birth Attendants | 216 | 23.9 |
| Accessibility and Infrastructure | 162 | 18.0 |
| Fear of Medical Interventions e.g. caesarean sections | 14 | 1.6 |
| Religious Beliefs | 81 | 9.0 |
| Influence of Family and Community | 85 | 9.4 |
| Trust and Relationship with Healthcare Providers | 83 | 9.2 |
| Perceived Severity of Labour | 26 | 2.9 |
| Influence of Family and Community | 34 | 3.8 |
| | **902** | **100.0** |

Out of the total 902 respondents, 619 women (68.6%) reported that they had delivered in a health facility at least once, while 283 women (31.4%) indicated that they had never delivered in a health facility. In terms of frequency of health facility use for delivery, 223 respondents (24.7%) had delivered once in a health facility. A further

200 women (22.2%) had delivered twice in such facilities, while 129 respondents (14.3%) had done so three times. The number of women who had delivered four times in health facilities was 48 (5.3%), and smaller proportions of 16 (1.8%) and 3 (0.3%) reported delivering five and six times, respectively. Notably, 283 women (31.4%) had never delivered in a health facility.

Looking specifically at the place of delivery preceding the last pregnancy, 360 respondents (39.9%) delivered in a private health facility, making this the most commonly reported setting. A slightly smaller group, 331 women (36.7%), had delivered in a public health facility. Meanwhile, 123 respondents (13.6%) reported giving birth at a religious centre, such as a church or mosque. Deliveries with a traditional birth attendant (TBA) were reported by 30 women (3.3%), and 56 respondents (6.2%) had delivered at home. A very small number, 2 women (0.2%), reported delivering on the way to a health facility.

Respondents were also asked to state the reasons for their choice of place of delivery preceding the last pregnancy. The most commonly cited reason was perceived quality of care, mentioned by 399 women (44.2%). This was followed by proximity to healthcare facilities, which influenced 224 respondents (24.8%), and the availability of skilled birth attendants, noted by 216 women (23.9%).

Other notable reasons included accessibility and infrastructure (162 respondents, 18.0%) and cost of delivery services, which was mentioned by 109 respondents (12.1%). A total of 92 women (10.2%) based their decision on previous birth experiences, while influence of family and community was reported by 85 respondents (9.4%).

Religious beliefs played a role for 81 respondents (9.0%), and trust and relationship with healthcare providers influenced the decision for 83 women (9.2%). Cultural beliefs and practices were cited by 69 respondents (7.7%), and 14 women (1.6%) reported that fear of medical interventions, such as caesarean sections, affected their choice. A smaller group of 26 respondents (2.9%) were influenced by the perceived severity of labour. It is noteworthy that influence of family and community was repeated in the data, reported by a total of 119 women (13.2%) if considered cumulatively.

**Table 8: Choice of the last Place of Delivery of reprductive age women (n = 902)**

| | Frequency (n) | Percentage (%) |
|---|---|---|
| **Place of delivery of the last pregnancy:** | | |
| A public health facility | 345 | 38.3 |
| A private health facility | 346 | 38.4 |
| A religious center (church, mosque) | 117 | 12.9 |
| With a Traditional Birth Attendant | 26 | 2.9 |
| At home | 67 | 7.4 |
| On the way to the health facility | 1 | 0.1 |
| **Reasons for the choice of last place of delivery:** | | |
| Proximity to Healthcare Facilities | 208 | 23.1 |
| Cost of Delivery Services | 117 | 13.0 |
| Cultural Beliefs and Practices | 70 | 7.8 |
| Perceived Quality of Care | 430 | 47.7 |
| Previous Birth Experiences | 131 | 14.5 |
| Availability of Skilled Birth Attendants | 231 | 25.6 |
| Accessibility and Infrastructure | 161 | 17.9 |
| Fear of Medical Interventions e.g. caesarean sections | 16 | 1.8 |
| Religious Beliefs | 73 | 8.1 |
| Influence of Family and Community | 70 | 7.8 |
| Trust and Relationship with Healthcare Providers | 81 | 9.0 |
| Perceived Severity of Labour | 22 | 2.4 |
| Influence of Family and Community | 35 | 3.9 |
| **Number of years since woman's last delivery: [5.1±4.7 years]** | | |
| 0-4 | 528 | 58.5 |
| 5-9 | 250 | 27.7 |
| 10-14 | 82 | 9.1 |
| 15-19 | 25 | 2.8 |
| 20-24 | 9 | 1.0 |
| 25-29 | 4 | 0.4 |
| 30-34 | 4 | 0.4 |
| | **902** | **100.0** |

Among the 902 reproductive age women surveyed, the most recent place of delivery varied across a range of settings. Private health facilities accounted for the highest proportion, with 346 women (38.4%) reporting that their last delivery took place in such a facility. This figure was closely followed by 345 respondents (38.3%) who delivered in public health facilities. Together, these figures show that more than three-

quarters of respondents (76.7%) delivered in a formal healthcare setting during their last pregnancy.

A total of 117 women (12.9%) reported that they gave birth in a religious centre, such as a church or mosque. Additionally, 26 respondents (2.9%) delivered with the assistance of a traditional birth attendant (TBA), while 67 women (7.4%) gave birth at home. Only one respondent (0.1%) reported delivering on the way to a health facility, suggesting that unplanned or emergency deliveries in transit were rare.

When asked about the reasons for choosing their last place of delivery, the most commonly cited factor was perceived quality of care, mentioned by 430 respondents (47.7%). This was followed by the availability of skilled birth attendants, which influenced 231 women (25.6%), and proximity to healthcare facilities, noted by 208 respondents (23.1%).

Other reasons included previous birth experiences, reported by 131 women (14.5%), and accessibility and infrastructure, mentioned by 161 respondents (17.9%). Cost was a determinant for 117 women (13.0%), while religious beliefs influenced the decision for 73 respondents (8.1%). Cultural beliefs and practices were cited by 70 women (7.8%), and the influence of family and community was also recorded by 70 women (7.8%), although it appeared again later in the table with 35 respondents (3.9%), suggesting a possible duplication or recoding in reporting.

Further reasons included trust and relationship with healthcare providers (81 respondents, 9.0%), fear of medical interventions such as caesarean sections (16 women, 1.8%), and the perceived severity of labour, reported by 22 respondents (2.4%). These findings suggest that both service quality and interpersonal factors play key roles in shaping delivery location choices.

The mean duration since the last delivery was 5.1 years, with a standard deviation of 4.7 years. More than half of the women, 528 respondents (58.5%), had delivered within the past 0–4 years. An additional 250 women (27.7%) reported delivering 5–9 years ago. The proportion declined progressively with time: 82 women (9.1%) had delivered 10–14 years prior, 25 respondents (2.8%) 15–19 years ago, and only small numbers of 9 (1.0%), 4 (0.4%), and 4 (0.4%) had delivered 20–24, 25–29, and 30–34 years ago respectively.

**History of Antenatal Visits of reproductive age women with childbirth experience**

**Table 9: History of antenatal visits (n = 902)**

| | Frequency (n) | Percentage (%) |
|---|---|---|
| **Booking status:** | | |
| Booked | 786 | 87.1 |
| Unbooked | 116 | 12.9 |
| **Reason for no ANC visit:** | | |
| Always busy at work | 6 | 0.7 |
| Always give birth at home | 1 | 0.1 |
| Because students are the one performi.. | 1 | 0.1 |
| Change of location | 1 | 0.1 |
| Cultural beliefs | 4 | 0.4 |
| Did not book at any health facility | 10 | 1.1 |
| Distance to nearest facility | 5 | 0.6 |
| Fear of being stigmatized | 1 | 0.1 |
| Financial constraints | 6 | 0.7 |
| Had ANC in a different facility in Og.. | 1 | 0.1 |
| Health worker | 1 | 0.1 |
| Husband is a doctor | 1 | 0.1 |
| I don't like hospitals at all | 1 | 0.1 |
| I don’t have interest in health facil.. | 1 | 0.1 |
| I usually go to church to drink anoin.. | 1 | 0.1 |
| I was booked in the mission | 1 | 0.1 |
| I was fine and baby was kicking well | 1 | 0.1 |
| I'm okay so Not interested for that | 1 | 0.1 |
| It's complicated | 1 | 0.1 |
| Late registration because of personal.. | 1 | 0.1 |
| Mother is a midwife | 1 | 0.1 |
| No problem | 3 | 0.3 |
| Not interested | 44 | 4.9 |
| Not interested am a busy woman | 1 | 0.1 |
| Not interested and I was strong | 1 | 0.1 |
| Not the first child and healthy | 1 | 0.1 |
| Pregnancy wasn’t planned | 1 | 0.1 |
| Previous experience of child birth | 1 | 0.1 |
| Religious belief | 6 | 0.7 |
| She had no health issues during pregn.. | 1 | 0.1 |
| She stayed in church during her pregn.. | 1 | 0.1 |
| She was not feeling fine | 1 | 0.1 |
| Time constraints | 4 | 0.4 |
| Travel | 2 | 0.2 |
| Scared of injection/drugs | 2 | 0.2 |
| *Booked* | *786* | *87.1* |
| **Pregnancy month of first ANC visit:** | | |
| 1-3 months | 415 | 46.0 |
| 4-6 months | 343 | 38.0 |
| 7-9 months | 28 | 3.1 |

| | | |
|---|---|---|
| *Unbooked* | *116* | *12.9* |
| **Number of total ANC visits during the last pregnancy:** | | |
| <6 visits | 98 | 10.9 |
| 6+ visits | 688 | 76.3 |
| *Unbooked* | *116* | *12.9* |
| **Sex of ANC provider:** | | |
| Male | 119 | 13.1 |
| Female | 667 | 74.0 |
| *Unbooked* | *116* | *12.9* |
| | **902** | **100.0** |

A majority of respondents, 786 women (87.1%), reported that they were booked for antenatal care (ANC) during their last pregnancy, whereas 116 women (12.9%) were unbooked, meaning they did not attend ANC services. Among those who did not attend ANC, a wide range of reasons were given. The most frequently cited reason was a general lack of interest, with 44 respondents (4.9%) explicitly stating they were not interested in attending ANC. Other reasons included being busy at work (6 respondents, 0.7%) and financial constraints (6 women, 0.7%). Some women expressed confidence in their pregnancy status, for example, saying “I was fine and baby was kicking well” or “I'm okay so not interested for that”, each cited by 1 respondent (0.1%). Others relied on religious practices, such as drinking anointing oil at church or staying in church during pregnancy. A few respondents mentioned practical barriers such as distance to the nearest facility (5 women, 0.6%) or travel (2 women, 0.2%).

Less commonly, reasons included fear of being stigmatised, cultural beliefs, and a previous experience of childbirth, which may have influenced a perception of reduced need for ANC. Some women mentioned specific, individual factors such as having a husband or mother who is a healthcare professional, or a dislike for hospitals. A very small number (less than 0.2% each) raised concerns about student healthcare

providers, complicated circumstances, or scared of injection or drugs. These diverse responses reflect both structural and personal factors influencing ANC attendance.

Regarding timing of the first ANC visit, 415 women (46.0%) booked in the first trimester (1–3 months), while 343 respondents (38.0%) initiated care during the second trimester (4–6 months). Only 28 women (3.1%) booked in the third trimester (7–9 months). The 116 unbooked respondents (12.9%) did not attend ANC at any point during their pregnancy.

With regard to the total number of ANC visits, 688 women (76.3%) reported attending six or more visits, in line with WHO recommendations. A further 98 women (10.9%) had fewer than six visits, and the remaining 116 (12.9%) were unbooked.

In terms of the sex of ANC provider, the majority of women, 667 respondents (74.0%), received care from female providers. A smaller group, 119 women (13.1%), were attended to by male providers, while the 116 unbooked women (12.9%) did not encounter any ANC provider during pregnancy.

**Women who delivered in health facilities (n = 619)**

**Table 10: Facilitating factors to choice of delivery at health facilities (n = 619)**

| | Frequency (n) | Percentage (%) |
|---|---|---|
| **Familiar with health providers in the health facility:** | | |
| Yes | 190 | 30.7 |
| No | 429 | 69.3 |
| **Basis of familiarity with health providers (n = 190):** | | |
| Family relations | 50 | 26.3 |
| Religious group member | 14 | 7.4 |
| Friend/peer | 57 | 30.0 |
| Business relations | 21 | 11.0 |
| Neighbours | 42 | 22.1 |
| Other types of familiarity | 6 | 3.2 |
| **Types of health workers attended to woman during labour:** | | |
| Doctor at PHC Facility | 307 | 49.6 |
| Nurse/midwife | 464 | 75.0 |
| Community Health Officer (CHO) | 125 | 20.2 |
| Community Health Extension Worker (CHEW) | 124 | 20.0 |
| **Healthcare provider who took the woman's delivery:** | | |
| Doctor | 263 | 42.5 |
| Nurse or Midwife | 211 | 34.1 |
| Community Health Officer (CHO) | 32 | 5.2 |
| Community Health Extension Worker (CHEW) | 97 | 15.7 |
| All the health workers at the health facility | 16 | 2.6 |
| | **619** | **100.0** |

Among the 619 women who delivered in health facilities, 190 respondents (30.7%) indicated that they were familiar with health providers in the facility where they delivered, while a greater proportion, 429 women (69.3%), reported no prior familiarity with the healthcare workers who attended to them. Among those who reported familiarity with providers (n = 190), the most commonly cited basis for familiarity was through friendship or peer relationships, reported by 57 women (30.0%). This was closely followed by family relations, reported by 50 women (26.3%), and neighbourhood interactions, which accounted for 42 respondents (22.1%). Other less common bases of familiarity included business relations (21

women, 11.0%), religious group membership (14 women, 7.4%), and other forms of familiarity, mentioned by 6 respondents (3.2%).

In terms of the type of health workers who attended to women during labour, nurses and midwives were the most frequently reported, mentioned by 464 respondents (75.0%). Doctors at primary health care (PHC) facilities attended to 307 women (49.6%), while community health officers (CHOs) and community health extension workers (CHEWs) were involved in the care of 125 (20.2%) and 124 (20.0%) women respectively. These findings indicate that multiple cadres of health professionals were involved in providing labour support, with nurses and midwives playing a central role.

Regarding the healthcare provider who took the actual delivery, the largest proportion of respondents, 263 women (42.5%), reported that their delivery was conducted by a doctor. A further 211 respondents (34.1%) were attended to by nurses or midwives. CHOs were responsible for deliveries in 32 cases (5.2%), while CHEWs assisted in 97 deliveries (15.7%). A small group of 16 women (2.6%) stated that all the health workers present participated in their delivery.

These results suggest that both the presence of skilled personnel and interpersonal connections with providers contributed to women's utilisation of health facilities for childbirth.

**To determine the factors influencing women's choice of maternal care in the region**

**Table 11: Percent distribution of reproductive age women 15-49 by who deliver last child at any health facilities and their choice of place of delivery of last child according to research area (n = 902)**

| | Ever delivered in Health Facility | | Place of Last Delivery[a] | | | | | | Number of women |
|---|---|---|---|---|---|---|---|---|---|
| Research area | Yes | No | Pub. HF | Priv. HF | RC | TBA | Home | In transit | |
| **Residence:** | | | | | | | | | |
| Urban | 69.7 | 30.3 | 41.0 | 40.4 | 12.0 | 2.4 | 4.0 | 0.3 | 376 |
| Rural | 67.9 | 32.1 | 36.3 | 36.9 | 13.7 | 3.2 | 10.0 | 0.0 | 526 |
| *P-value* | *0.564* | | | | ***0.012*** | | | | |
| **Zone:** | | | | | | | | | |
| Osun Central | 67.0 | 33.0 | 34.2 | 37.0 | 14.6 | 3.6 | 10.6 | 0.0 | 451 |
| Osun East | 70.3 | 29.7 | 42.4 | 39.7 | 11.3 | 2.2 | 4.2 | 0.2 | 451 |
| *P-value* | *0.282* | | | | ***0.001*** | | | | |
| **LGA:** | | | | | | | | | |
| Boluwaduro | 68.9 | 31.1 | 31.1 | 37.8 | 11.1 | 4.4 | 15.6 | 0.0 | 45 |
| Ifelodun | 69.6 | 30.4 | 39.1 | 37.0 | 10.9 | 6.5 | 6.5 | 0.0 | 46 |
| Odo Otin | 76.1 | 23.9 | 39.1 | 37.0 | 8.7 | 2.2 | 13.0 | 0.0 | 46 |
| Boripe | 53.3 | 46.7 | 24.4 | 33.3 | 26.7 | 4.4 | 11.1 | 0.0 | 45 |
| Ila | 57.8 | 42.2 | 22.2 | 44.4 | 15.6 | 2.2 | 15.6 | 0.0 | 45 |
| Ifedayo | 70.8 | 29.2 | 39.6 | 29.2 | 22.9 | 0.0 | 8.3 | 0.0 | 48 |
| Irepodun | 61.7 | 38.3 | 36.2 | 38.3 | 14.9 | 2.1 | 8.5 | 0.0 | 47 |
| Olorunda | 62.2 | 37.8 | 37.8 | 35.6 | 17.8 | 6.7 | 2.2 | 0.0 | 45 |
| Osogbo | 70.7 | 29.3 | 43.1 | 39.7 | 10.3 | 5.2 | 1.7 | 0.0 | 58 |
| Atakunmosa East | 75.6 | 24.4 | 37.8 | 40.0 | 11.1 | 4.4 | 6.7 | 0.0 | 45 |
| Atakunmosa West | 67.4 | 32.6 | 32.6 | 41.3 | 10.9 | 6.5 | 8.7 | 0.0 | 46 |
| Ife East | 67.3 | 32.7 | 41.8 | 36.4 | 14.6 | 0.0 | 7.3 | 0.0 | 55 |
| Ife North | 68.3 | 31.7 | 390 | 34.2 | 12.2 | 2.4 | 12.2 | 0.0 | 41 |
| Ife South | 66.0 | 34.0 | 38.0 | 46.0 | 14.0 | 2.0 | 0.0 | 0.0 | 50 |
| Ife Central | 75.0 | 25.0 | 37.5 | 48.2 | 8.9 | 1.8 | 3.6 | 0.0 | 56 |
| Obokun | 71.4 | 28.6 | 42.9 | 35.7 | 11.9 | 2.4 | 7.1 | 0.0 | 42 |
| Oriade | 76.7 | 23.3 | 60.0 | 33.3 | 3.3 | 0.0 | 3.3 | 0.0 | 30 |
| Ilesa East | 78.7 | 21.3 | 54.1 | 36.1 | 6.6 | 1.6 | 1.6 | 0.0 | 61 |
| Ilesa West | 64.7 | 35.3 | 31.4 | 41.2 | 13.7 | 0.0 | 11.8 | 2.0 | 51 |
| *P-value* | *0.504* | | | | *0.312* | | | | |
| **Total** | **619 (68.6)** | **283 (31.4)** | **345 (38.3)** | **346 (38.4)** | **117 (13.0)** | **26 (2.9)** | **67 (7.4)** | **1 (0.1)** | **902 (100.0)** |

*a. Pub. HF = Public Health Facility; Priv. HF = Private Health Facility; RC = Religious Centre; TBA = Traditional Birth Attendants; Home = Home delivery; In transit = Woman delivers on the road to health facility.*

Out of the 902 reproductive age women surveyed, 68.6% (619 women) reported that they had ever delivered in a health facility, while 31.4% (283 women) had not. This reflects relatively high utilisation of facility-based delivery across the state.

When disaggregated by residence, 69.7% of urban women and 67.9% of rural women had ever delivered in a health facility, showing only a marginal difference. However, a more noticeable difference was observed in the place of last delivery. Among urban residents, 41.0% delivered in public health facilities, and 40.4% in private facilities, whereas in rural areas, 36.3% used public health facilities and 36.9% used private ones. Deliveries at religious centres were slightly more common in rural areas (13.7%) compared to urban (12.0%). Notably, home births were significantly more frequent in rural settings (10.0%) than urban areas (4.0%). Deliveries with traditional birth attendants (TBAs) were reported by 2.4% of urban women and 3.2% of rural women. A small number of urban respondents (0.3%) delivered in transit, while this was not reported in rural areas. While the difference in facility-based delivery between urban and rural areas was not statistically significant ($p = 0.564$), the variation in place of last delivery was significant ($p = 0.012$).

Across zones, 67.0% of women in Osun Central and 70.3% in Osun East had ever delivered in a health facility. However, their place of last delivery differed significantly ($p = 0.001$). Women in Osun East were more likely to use public health facilities (42.4%) than those in Osun Central (34.2%). Similarly, use of private health facilities was slightly higher in Osun East (39.7%) compared to Osun Central (37.0%). Deliveries at religious centres were more frequent in Osun Central (14.6%) than in Osun East (11.3%), and home births were substantially more common in Osun Central (10.6%) versus Osun East (4.2%). Deliveries with TBAs were also slightly more frequent in Osun Central (3.6%) than Osun East (2.2%). Only one case of delivery in transit was recorded in Osun East.

Analysis by Local Government Area (LGA) revealed considerable variation. The proportion of women who had ever delivered in a health facility ranged from 53.3%

in Boripe to 78.7% in Ilesa East, indicating both geographical disparities and differences in healthcare utilisation across LGAs.

In terms of place of last delivery, public health facility use was highest in Oriade (60.0%), Obokun (42.9%), and Atakunmosa East (37.8%), while private health facilities were more commonly used in LGAs like Ilesa East (54.1%) and Ife Central (48.2%). Delivery at religious centres was most reported in Boripe (26.7%) and Ifedayo (22.9%). Home births were notably high in Boluwaduro (15.6%), Ila (15.6%), and Odo Otin (13.0%). TBAs were mostly used in Olorunda (6.7%), Osogbo (5.2%), and Irepodun (2.1%). Deliveries in transit were reported only in Ilesa West (2.0%), though this was rare overall.

Despite observable differences across LGAs, the associations between LGA and either having ever delivered in a facility ($p = 0.504$) or the place of last delivery ($p = 0.312$) were not statistically significant at the 5% level.

**Table 12: Percent distribution of reproductive age women 15-49 by who deliver last child at any health facilities and their choice of place of delivery of last child according to women's background characteristics (n = 902)**

| Background characteristics | Ever delivered in Health Facility | | Place of Last Delivery[a] | | | | | | Number of women |
|---|---|---|---|---|---|---|---|---|---|
| | Yes | No | Pub. HF | Priv. HF | RC | TBA | Home | In transit | |
| **Age:** | | | | | | | | | |
| 15 – 19 years | 80.0 | 20.0 | 40.0 | 20.0 | 40.0 | 0.0 | 0.0 | 0.0 | 5 |
| 20 – 24 years | 77.0 | 23.0 | 42.6 | 34.4 | 8.2 | 8.2 | 6.6 | 0.0 | 61 |
| 25 – 29 years | 59.1 | 40.9 | 36.4 | 38.1 | 17.6 | 2.8 | 5.1 | 0.0 | 176 |
| 30 – 34 years | 65.6 | 34.4 | 37.6 | 40.8 | 11.9 | 1.4 | 8.3 | 0.0 | 218 |
| 35 – 39 years | 76.1 | 23.9 | 42.6 | 39.1 | 10.7 | 1.0 | 6.1 | 0.5 | 197 |
| 40 – 44 years | 73.2 | 26.8 | 33.3 | 43.8 | 10.5 | 5.2 | 7.2 | 0.0 | 153 |
| 45 – 49 years | 64.1 | 35.9 | 39.1 | 26.1 | 17.4 | 3.3 | 14.1 | 0.0 | 92 |
| *P-value* | ***0.006*** | | | | *0.059* | | | | |
| **Formal education:** | | | | | | | | | |
| No formal education | 46.6 | 53.4 | 31.6 | 29.5 | 19.2 | 5.7 | 14.0 | 0.0 | 193 |
| Had formal education | 74.6 | 25.4 | 40.1 | 40.8 | 11.3 | 2.1 | 5.6 | 0.1 | 709 |
| | ***<0.0001*** | | | | ***<0.0001*** | | | | |
| **Marital Status:** | | | | | | | | | |
| Married | 68.5 | 31.5 | 38.2 | 38.5 | 12.9 | 2.7 | 7.6 | 0.1 | 867 |
| Single | 73.1 | 26.9 | 42.3 | 30.8 | 19.2 | 7.7 | 0.0 | 0.0 | 26 |
| Divorced | 66.7 | 33.3 | 33.3 | 44.4 | 0.0 | 11.1 | 11.1 | 0.0 | 9 |
| | *0.878* | | | | *0.519* | | | | |
| **Ethnicity:** | | | | | | | | | |
| Yoruba | 69.6 | 30.4 | 39.3 | 38.4 | 13.8 | 2.7 | 5.7 | 0.1 | 822 |
| Ibo | 62.0 | 38.0 | 28.0 | 42.0 | 8.0 | 6.0 | 16.0 | 0.0 | 50 |
| Hausa | 42.1 | 57.9 | 21.1 | 26.3 | 0.0 | 0.0 | 52.6 | 0.0 | 19 |
| Others | 72.7 | 27.3 | 36.4 | 36.4 | 0.0 | 9.1 | 18.2 | 0.0 | 11 |
| | *0.053* | | | | ***<0.0001*** | | | | |
| **Husband's occupation:** | | | | | | | | | |
| Unemployed | 67.7 | 32.4 | 32.4 | 35.3 | 17.7 | 8.8 | 5.9 | 0.0 | 34 |
| Self-employed/Entrepr. | 65.0 | 35.0 | 37.5 | 36.7 | 13.9 | 3.6 | 8.6 | 0.0 | 592 |
| Private/Professional | 73.6 | 26.4 | 36.4 | 43.4 | 14.7 | 1.6 | 3.1 | 0.8 | 129 |
| Civil Servant | 78.9 | 21.1 | 44.2 | 41.5 | 7.5 | 0.0 | 6.8 | 0.0 | 147 |
| | ***0.006*** | | | | ***0.017*** | | | | |
| **Ownership of means of transportation:** | | | | | | | | | 622 |
| Yes | 69.1 | 30.9 | 37.5 | 41.0 | 11.9 | 2.3 | 7.2 | 0.2 | 280 |
| No | 67.5 | 32.5 | 40.0 | 32.5 | 15.4 | 4.3 | 7.9 | 0.0 | |
| *P-value* | *0.625* | | | | *0.106* | | | | |
| **Total** | **619 (68.6)** | **283 (31.4)** | **345 (38.3)** | **346 (38.4)** | **117 (13.0)** | **26 (2.9)** | **67 (7.4)** | **1 (0.1)** | **902 (100.0)** |

*a. Pub. HF = Public Health Facility; Priv. HF = Private Health Facility; RC = Religious Centre; TBA = Traditional Birth Attendants; Home = Home delivery; In transit = Woman delivers on the road to health facility.*

Out of all women surveyed by age group, the highest proportion of facility-based delivery was observed among women aged 15–19 years (80.0%) and 20–24 years

(77.0%), while women aged 25–29 years reported the lowest facility-based delivery rate at 59.1%. Despite relatively high facility use among younger women, home births remained notable in the 15–19 years (40.0%) and 45–49 years (17.4%) groups. Use of religious centres declined with age but remained notable across all age brackets. The association between age and ever delivering in a health facility was statistically significant ($p = 0.006$), although the association with place of last delivery approached significance ($p = 0.059$).

Formal education appeared to be a strong determinant of health facility use. Among women with no formal education, only 46.6% had ever delivered in a health facility, compared to 74.6% among those with formal education. This relationship was statistically significant ($p < 0.0001$). Place of last delivery also varied: those with formal education were more likely to deliver in private health facilities (40.8%) and less likely to give birth at home (5.6%) or with TBAs (2.1%) than women without formal education. In contrast, women without formal education had higher rates of home births (14.0%), TBA use (5.7%), and deliveries in religious centres (19.2%). The variation in place of last delivery by education was also statistically significant ($p < 0.0001$).

Analysis by marital status revealed little variation in health facility delivery, with married women (68.5%), single women (73.1%), and divorced women (66.7%) reporting similar rates of ever delivering in a health facility. No statistically significant association was found between marital status and either outcome ($p = 0.878$ for facility delivery; $p = 0.519$ for place of last delivery).

With respect to ethnicity, the highest proportion of facility-based delivery was observed among Yoruba (69.6%) and "Other" ethnic groups (72.7%), while Hausa

women reported the lowest rate (42.1%). Facility-based delivery was significantly associated with ethnicity ($p = 0.053$), and the place of last delivery showed marked differences ($p < 0.0001$). For instance, Hausa women were far more likely to report home births (52.6%) than Yoruba (5.7%) or Ibo women (16.0%). Deliveries in religious centres were more common among Yoruba (13.8%) than Ibo (8.0%).

The occupation of the husband was significantly associated with both facility delivery ($p = 0.006$) and place of last delivery ($p = 0.017$). Women whose husbands were civil servants reported the highest rate of health facility delivery at 78.9%, while those whose husbands were self-employed had the lowest at 65.0%. Deliveries in public health facilities were most common among women married to civil servants (44.2%), while private facility use was highest among those with husbands in private/professional occupations (43.4%). Home deliveries and use of TBAs were lowest among civil servants' wives.

Regarding ownership of a means of transportation, no statistically significant association was found with either health facility use ($p = 0.625$) or place of last delivery ($p = 0.106$). However, women who had access to transportation reported slightly higher facility-based delivery rates (69.1%) than those who did not (67.5%). Private facility use was marginally more common among those with transportation (41.0%) than among those without (32.5%), while home births were slightly more prevalent among women without transport access (7.9%) compared to those with transport (7.2%).

**Table 13: Percent distribution of reproductive age women 15-49 by who deliver last child at any health facilities and their choice of place of delivery of last child according to obstetrics profile (n = 902)**

| Obstetrics profile | Ever delivered in Health Facility | | Place of Last Delivery[a] | | | | | | Number of women |
|---|---|---|---|---|---|---|---|---|---|
| | Yes | No | Pub. HF | Priv. HF | RC | TBA | Home | In transit | |
| **Age at first birth:** | | | | | | | | | |
| 15 – 19 years | 56.3 | 43.7 | 27.6 | 34.5 | 19.5 | 2.3 | 16.1 | 0.0 | 87 |
| 20 – 24 years | 66.7 | 33.3 | 39.1 | 37.8 | 11.3 | 4.7 | 7.1 | 0.0 | 381 |
| 25 – 29 years | 68.6 | 31.4 | 36.7 | 39.0 | 15.3 | 1.8 | 7.0 | 0.3 | 341 |
| 30 – 34 years | 88.1 | 11.9 | 52.4 | 40.5 | 6.0 | 0.0 | 1.2 | 0.0 | 84 |
| 35 – 39 years | 88.9 | 11.1 | 33.3 | 55.6 | 0.0 | 0.0 | 11.1 | 0.0 | 9 |
| *P-value* | ***<0.0001*** | | | | ***0.003*** | | | | |
| **Gravidity:** | | | | | | | | | |
| Primigravida | 66.5 | 33.5 | 39.4 | 41.4 | 12.3 | 2.5 | 4.4 | 0.0 | 203 |
| Multigravida | 69.2 | 30.8 | 37.9 | 37.5 | 13.2 | 3.0 | 8.3 | 0.1 | 699 |
| | *0.459* | | | | *0.491* | | | | |
| **Parity** | | | | | | | | | |
| Primiparous | 64.8 | 35.2 | 39.8 | 41.0 | 13.3 | 2.3 | 3.5 | 0.0 | 256 |
| Multiparous | 70.1 | 29.9 | 37.6 | 37.3 | 12.9 | 3.1 | 9.0 | 0.2 | 646 |
| | *0.123* | | | | *0.107* | | | | |
| **Number of children alive:** | | | | | | | | | |
| 0 | 66.7 | 33.3 | 33.3 | 66.7 | 0.0 | 0.0 | 0.0 | 0.0 | 0.3 |
| 1-4 | 69.1 | 30.9 | 39.1 | 39.2 | 12.7 | 2.8 | 6.2 | 0.1 | 92.8 |
| 5-8 | 61.9 | 37.1 | 27.4 | 25.8 | 17.7 | 4.8 | 24.2 | 0.0 | 6.9 |
| | *0.600* | | | | ***<0.0001*** | | | | |
| **Birth Order of the Last Child:** | | | | | | | | | |
| 0 | 75.0 | 25.0 | 75.0 | 25.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.4 |
| 1 | 66.0 | 34.0 | 39.8 | 40.9 | 13.1 | 2.3 | 3.9 | 0.0 | 28.7 |
| 2-3 | 70.6 | 29.4 | 38.8 | 39.7 | 11.8 | 2.2 | 7.4 | 0.2 | 50.9 |
| 4-5 | 67.5 | 32.5 | 35.7 | 33.8 | 15.6 | 5.2 | 9.7 | 0.0 | 17.1 |
| 6+ | 65.4 | 34.6 | 23.1 | 19.2 | 19.2 | 7.7 | 30.8 | 0.0 | 2.9 |
| | *0.750* | | | | ***0.003*** | | | | |
| **Adverse birth outcomes during delivery preceding the last pregnancy:** | | | | | | | | | 12.1 |
| Yes | 84.4 | 15.6 | 41.3 | 44.0 | 9.2 | 1.8 | 2.8 | 0.9 | 87.9 |
| No | 66.5 | 33.5 | 37.8 | 37.6 | 13.5 | 3.0 | 8.1 | 0.0 | |
| | ***<0.0001*** | | | | ***0.015*** | | | | |
| **Booking status:** | | | | | | | | | |
| Booked | 74.1 | 25.9 | 41.4 | 42.6 | 11.1 | 1.3 | 3.6 | 0.1 | 13.2 |
| Unbooked | 31.9 | 68.1 | 17.2 | 9.5 | 25.9 | 13.8 | 33.6 | 0.0 | 86.8 |
| | ***<0.0001*** | | | | ***<0.0001*** | | | | |
| **Pregnancy related problem during the last pregnancy:** | | | | | | | | | 13.2<br>86.8 |
| Yes | 85.7 | 14.3 | 37.0 | 46.2 | 7.6 | 2.5 | 6.7 | 0.0 | |
| No | 66.0 | 34.0 | 38.4 | 37.2 | 13.8 | 2.9 | 7.5 | 0.1 | |
| | ***<0.0001*** | | | | *0.340* | | | | |
| **Outcomer of the last pregnancy:** | | | | | | | | | 98.3 |
| Live birth | 68.7 | 31.3 | 38.1 | 38.3 | 13.2 | 2.7 | 7.6 | 0.1 | 1.11 |
| Stilll-birth | 70.0 | 30.0 | 40.0 | 50.0 | 0.0 | 10.0 | 0.0 | 0.0 | 0.6 |
| Died few days after | 60.0 | 40.0 | 60.0 | 20.0 | 0.0 | 20.0 | 0.0 | 0.0 | |
| *P-value* | *0.913* | | | | *0.321* | | | | |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Total** | **68.6** | **31.4** | **345 (38.3)** | **346 (38.4)** | **117 (13.0)** | **26 (2.9)** | **67 (7.4)** | **1 (0.1)** | **902 (100.0)** |

*a. Pub. HF = Public Health Facility; Priv. HF = Private Health Facility; RC = Religious Centre; TBA = Traditional Birth Attendants; Home = Home delivery; In transit = Woman delivers on the road to health facility.*

Age at First Birth

The proportion of women who had ever delivered in a health facility increased with age at first birth. Among those who gave birth for the first time between 15–19 years, only 56.3% had used a health facility. In contrast, those whose first birth occurred at ages 30–34 and 35–39 reported much higher facility use, at 88.1% and 88.9%, respectively. This association was highly significant ($p < 0.0001$).

Regarding place of last delivery, private health facility use was most common among women whose first birth occurred at age 35–39 years (55.6%). Public health facility use was highest for the 30–34 age group (52.4%), while home births and use of religious centres were more common among those who started childbearing before age 25. The association between age at first birth and place of last delivery was statistically significant ($p = 0.003$).

Gravidity and Parity

There were no statistically significant differences in the likelihood of facility-based delivery by gravidity ($p = 0.459$) or parity ($p = 0.123$). Nonetheless, multiparous women were slightly more likely to have delivered in a health facility (70.1%) compared to primiparous women (64.8%).

A similar trend was observed in the place of last delivery. Use of private health facilities was slightly more common among primigravida and primiparous women, while home births were notably higher among multiparous and multigravid women.

### Number of Living Children

There was no significant association between the number of living children and having ever delivered in a health facility ($p = 0.600$). However, the place of last delivery varied significantly ($p < 0.0001$). Among women with 5–8 living children, 24.2% had delivered at home, and only 27.4% had used public health facilities. In contrast, women with no children alive all delivered in either public or private facilities.

### Birth Order of Last Child

The association between birth order and ever delivering in a facility was not significant ($p = 0.750$), but the place of last delivery differed significantly ($p = 0.003$). Women whose last child was of higher birth order (6 or more) were more likely to have delivered at home (30.8%) or with TBAs (7.7%), and less likely to have used public health facilities (23.1%).

### Adverse Birth Outcomes (Previous Pregnancy)

Women who had experienced adverse birth outcomes during the delivery preceding the last pregnancy were significantly more likely to have delivered their last child in a health facility (84.4%), compared to 66.5% among those with no such history ($p < 0.0001$).

While use of public and private facilities was comparable between groups, home births were substantially lower among those with adverse birth histories (2.8%) than those without (8.1%) ($p = 0.015$).

### Antenatal Care Booking Status

Booking status was strongly associated with place of delivery ($p < 0.0001$). Among booked women, 74.1% delivered in a health facility, compared to only 31.9% of those unbooked. Booked women were more likely to use both public (41.4%) and private (42.6%) facilities. In contrast, unbooked women had higher rates of home births (33.6%), religious centre deliveries (25.9%), and use of TBAs (13.8%).

Pregnancy-Related Problems (Last Pregnancy)

Women who reported pregnancy-related complications were more likely to deliver in a health facility (85.7%) than those without complications (66.0%), a statistically significant difference ($p < 0.0001$). However, their place of last delivery did not significantly differ ($p = 0.340$), suggesting that complications influenced the decision to use a facility, but not necessarily the type.

Outcome of Last Pregnancy

There was no significant association between birth outcome (live birth, stillbirth, or early neonatal death) and either ever delivering in a health facility ($p = 0.913$) or place of last delivery ($p = 0.321$). However, the small number of cases of stillbirth and early neonatal death may have limited the statistical power.

**Table 14: Bivariate multilevel logistic regression analysis for individual and community level factors influencing women's choice of place of delivery among reproductive age women 15-49 who deliver last child at any health facilities (n = 902)**

| Predictors[a] | COR | P-value[c] | 95% CI | ICC | LR Test (p) |
|---|---|---|---|---|---|
| *Individual level factors* | | | | | |
| **Age:** | | | | | |
| 15 – 19 years | Ref | - | - | 0.026 | 3.33 |
| 20 – 24 years | 2.053 | 0.461 | 0.30 - 13.92 | | (0.0341) |
| 25 – 29 years | 1.794 | 0.536 | 0.28 - 11.40 | | |
| 30 – 34 years | 2.299 | 0.376 | 0.36 - 14.51 | | |
| 35 – 39 years | 2.776 | 0.279 | 0.43 - 17.65 | | |
| 40 – 44 years | 2.077 | 0439 | 0.32 - 13.24 | | |
| 45 – 49 years | 1.151 | 0.882 | 0.17 - 7.45 | | |
| **Formal education:** | | | | | |
| No formal education | Ref | - | - | 0.0037 | 0.08 |
| Had formal education | 2.657 | **<0.0001** | 1.86 - 3.77 | | (0.3870) |
| **Highest completed level of education:** | | | | | |
| Primary | Ref | **-** | - | 0.0037 | 0.04 |
| Secondary | 2.280 | **0.010** | 1.22 - 4.25 | | (0.4172) |
| Post-secondary | 5.028 | **<0.0001** | 2.52 - 10.0 | | |
| Post-tertiary | 17.19 | **<0.0001** | 3.74 - 79.0 | | |
| **Marital Status:** | | | | | |
| Married | Ref | - | - | 0.026 | 3.37 |
| Single | 0.819 | 0.663 | 0.33 - 2.00 | | (0.0333) |
| Divorced | 1.228 | 0.801 | 0.24 - 6.10 | | |
| **Ethnicity:** | | | | | |
| Yoruba | Ref | - | - | 0.520 | 3.49 |
| Ibo | 0.656 | 0.194 | 0.34 - 1.23 | | (0.0308) |
| Hausa | 0.249 | **0.004** | 0.09 - 0.63 | | |
| Others | 0.700 | 0.609 | 0.17 - 2.74 | | |
| **Husband's occupation:** | | | | | |
| Unemployed | Ref | - | - | 0.020 | 2.05 |
| Self-employed/Entrepr. | 1.357 | 0.425 | 0.64 - 2.87 | | (0.0762) |
| Private/Professional | 1.859 | 0.151 | 0.79 - 4.33 | | |
| Civil Servant | 2.752 | **0.022** | 1.15 - 6.53 | | |
| **Ownership of means of transportation:** | | | | | |
| Yes | Ref | - | - | 0.028 | 3.86 |
| No | 0.701 | **0.037** | 0.50 - 0.97 | | (0.0247) |
| **Age at first birth:** | | | | | |
| 15 – 19 years | Ref | - | - | 0.0160 | 1.34 |
| 20 – 24 years | 2.023 | **0.006** | 1.22 - 3.33 | | (0.1232 |
| 25 – 29 years | 1.862 | **0.016** | 1.12 - 3.09 | | |
| 30 – 34 years | 7.582 | **<0.0001** | 2.94 - 19.5 | | |
| 35 – 39 years | 4.792 | 0.150 | 0.56 - 40.4 | | |
| **Gravidity:** | | | | | |
| Primigravida | Ref | - | - | 0.024 | 3.15 |
| Multigravida | 0.731 | 0.121 | 0.49 - 1.08 | | (0.0380) |
| **Parity** | | | | | |
| Primiparous | Ref | - | - | 0.023 | 2.82 |
| Multiparous | 0.719 | 0.076 | 0.49 - 1.03 | | (0.0465) |
| **Number of children alive:** | | | | | |
| 0 | Ref | - | - | 0.026 | 3.21 |
| 1-4 | 3.21 | **<0.0001** | 1.88 - 5.48 | | (0.0366) |
| 5-8 | 1[b] | | | | |
| **Birth Order of the Last Child:** | | | | | |

| | | | | | |
|---|---|---|---|---|---|
| 0 | Ref | - | - | 0.185 | 2.27 |
| 1 | 5.634 | **<0.0001** | 2.40 - 13.1 | | (0.0660) |
| 2-3 | 4.984 | **<0.0001** | 2.19 - 11.3 | | |
| 4-5 | 3.094 | **0.010** | 1.30 - 7.32 | | |
| 6+ | 1[b] | | | | |
| **Adverse birth outcomes during delivery preceding the last pregnancy:** | | | | 0.029 | 4.18 |
| Yes | Ref | - | - | | (0.0204) |
| No | 0.498 | **0.015** | 0.28 - 0.87 | | |
| **Booking status:** | | | | | |
| Booked | Ref | - | - | 0.019 | 1.47 |
| Unbooked | 0.068 | **<0.0001** | 0.04 - 0.10 | | (0.1128) |
| **Pregnancy related problem during the last pregnancy:** | | | | | |
| Yes | Ref | - | - | 0.029 | 4.01 |
| No | 0.593 | **0.048** | 0.35 - 0.99 | | (0.0226) |
| **Outcome of the last pregnancy:** | | | | | |
| Live birth | Ref | - | - | 0.026 | 3.54 |
| Stilll-birth | 3.094 | 0.289 | 0.38 - 24.97 | | (0.0299) |
| Died few days after | 1.029 | 0.979 | 0.11 - 9.55 | | |
| *Community level fators* | | | | | |
| **Residence:** | | | | | |
| Urban | Ref | - | - | 0.0074 | 0.30 |
| Rural | 0.628 | **0.011** | 0.44 - 0.89 | | (0.2905) |
| **Zone:** | | | | | |
| Osun Central | Ref | - | - | 0 | 0.00 |
| Osun East | 1.849 | **<0.0001** | 1.34 - 2.53 | | *na*[b] |

*a. This model estimates the odds of choosing health facility as the choice of place of delivery over other possible choices.*

b. *Omitted*

c. *P-value is significant at 0.05*

The bivariate multilevel logistic regression analysis was conducted to identify individual and community-level factors associated with women's choice of place of delivery among women aged 15 to 49 years who delivered their most recent child. This approach accounted for the nested structure of the data by introducing a random intercept at the cluster level (Local Government Area - LGA). The random intercept models tested whether variability in the odds of facility delivery could be attributed to differences between LGAs. The dependent variable was the place of delivery for the last birth, coded as a binary outcome where 1 indicated choice of place of delivery at a health facility and 0 indicated choice of place of delivery elsewhere. Crude odds ratios (CORs) were estimated to quantify the likelihood of choice of facility delivery

associated with each predictor. Model fit was evaluated using the likelihood ratio (LR) test, while intra-class correlation (ICC) measured the proportion of variance attributable to clustering at the LGA level.

The analysis began with maternal age group, with women aged 15–19 years serving as the reference category. Although all age groups showed higher crude odds ratios (CORs) for facility delivery compared to the reference, none of the associations reached statistical significance at the 5% level. Nonetheless, the likelihood ratio (LR) test for the random effect of LGA was statistically significant ($p = 0.0341$), and the intraclass correlation coefficient (ICC) was 0.026, indicating that 2.6% of the variation in delivery place choice was due to clustering at the LGA level. As the overall p-value for age group was above 0.25, this variable was retained for further analysis in the multivariable model.

Maternal formal education showed a statistically significant association with facility delivery. Women with formal education were significantly more likely to deliver in health facilities compared to those without formal education (COR = 2.657; 95% CI: 1.86–3.77; $p < 0.0001$). However, the LR test for the model did not suggest significant random effects from clustering ($p = 0.3870$), and the ICC was low (0.0037).

In terms of highest level of education attained, a gradient effect was observed. Compared to women with only primary education, those with secondary (COR = 2.280; $p = 0.010$), post-secondary (COR = 5.028; $p < 0.0001$), and post-tertiary education (COR = 17.19; $p < 0.0001$) were progressively more likely to utilise facility-based delivery services. Although the model's ICC was again low at 0.0037,

the consistent trend across educational levels justifies inclusion in multivariable modelling.

Marital status did not show significant associations, with single (COR = 0.819; p = 0.663) and divorced women (COR = 1.228; p = 0.801) not significantly differing from married women in their odds of facility delivery. However, the model had a modest ICC of 0.026 and significant random effects (LR test p = 0.0333), warranting consideration of this variable in subsequent models.

Ethnicity was significantly associated with facility delivery. Compared to Yoruba women, those of Ibo ethnicity had significantly lower odds (COR = 0.249; 95% CI: 0.09–0.63; p = 0.004), while other ethnic groups showed no statistically significant differences. The ICC was 0.520 and the LR test was statistically significant (p = 0.0308), suggesting notable between-cluster variability.

Regarding the occupation of the husband, women whose husbands were civil servants had significantly higher odds of delivering at a health facility (COR = 2.752; 95% CI: 1.15–6.53; p = 0.022), while the other occupational categories showed non-significant associations. The ICC for this model was 0.020 with marginally significant clustering effects (LR test p = 0.0762).

Ownership of a means of transportation was associated with lower odds of facility delivery (COR = 0.701; 95% CI: 0.50–0.97; p = 0.037), indicating that women without transportation were less likely to deliver at health facilities. The ICC was 0.028, and random effect significance was confirmed (p = 0.0247), suggesting the importance of both individual- and community-level effects.

Age at first birth emerged as a strong predictor. Compared to women who had their first birth at ages 15–19, those who began childbearing at ages 20–24 (COR = 2.023; p = 0.006), 25–29 (COR = 1.862; p = 0.016), and 30–34 (COR = 7.582; p < 0.0001) were more likely to deliver at health facilities. Women aged 35–39 at first birth also had increased odds (COR = 4.792), though this association was not statistically significant (p = 0.150). The ICC was 0.016 and the LR test was not significant (p = 0.1232), suggesting weak clustering effects.

Gravidity and parity were both marginally associated with facility delivery. Multigravida women were less likely to deliver at a facility compared to primigravida women (COR = 0.731; p = 0.121), while multiparous women had lower odds compared to primiparous women (COR = 0.719; p = 0.076). These associations, though not statistically significant at the 5% level, were below the 0.25 threshold and therefore retained. ICCs for both were low (0.024 and 0.023 respectively), but random effects were significant (LR p-values = 0.0380 and 0.0465 respectively).

The number of children alive was significantly associated with delivery place. Women with 1–4 children had over three times higher odds of facility delivery compared to those with none (COR = 3.21; 95% CI: 1.88–5.48; p < 0.0001). The ICC was 0.026 and LR test was significant (p = 0.0366).

Birth order of the last child also demonstrated a graded association. Compared to women whose last child was their firstborn, those with higher birth orders (2–3 and 4–5) had significantly lower odds of facility delivery, with decreasing odds at higher birth orders. All categories, except the 6+ group, were statistically significant. The ICC was 0.185, and random intercept significance was borderline (p = 0.0660), justifying its inclusion.

Adverse birth outcomes during the delivery preceding the last pregnancy were associated with reduced odds of facility delivery in the current pregnancy (COR = 0.498; 95% CI: 0.28–0.87; p = 0.015). The ICC for this model was 0.029 with significant random effects (p = 0.0204), highlighting community-level variation.

Booking status for antenatal care was a strong predictor. Unbooked women were significantly less likely to deliver in a health facility (COR = 0.068; 95% CI: 0.04–0.10; p < 0.0001). The ICC was 0.019 with non-significant random effects (p = 0.1128), though the strength of association supports its inclusion in multivariable analysis.

Having no pregnancy-related complications was associated with lower odds of delivering in a facility compared to those who experienced complications (COR = 0.593; 95% CI: 0.35–0.99; p = 0.048). The ICC was 0.029 and the LR test was statistically significant (p = 0.0226), affirming relevance of community context.

Finally, the outcome of the last pregnancy did not show a significant association with facility delivery. Compared to women who had a live birth, those with stillbirth or neonatal death had higher odds of facility delivery, though these were not statistically significant. Nonetheless, the ICC was 0.026 with a significant LR test (p = 0.0299), supporting its inclusion for further multilevel analysis.

Among the community-level variables, residence and zone were included in the bivariate analysis. Compared to urban dwellers, rural women had significantly lower odds of delivering at a health facility (COR = 0.628, 95% CI: 0.44–0.89, p = 0.011), suggesting a substantial disparity in institutional delivery by place of residence. The

ICC for this model was 0.0074, indicating that less than 1% of the variance in the choosing health facility delivery was attributable to differences between LGAs. The likelihood ratio test for this model did not reach statistical significance ($p = 0.2905$), suggesting limited random intercept variation after adjusting for residence alone.

Geopolitical zone also showed a statistically significant association with place of delivery. Compared to women residing in Osun Central, those in Osun East had significantly higher odds of delivering at a health facility (COR = 1.849, 95% CI: 1.34–2.53, $p < 0.0001$). The ICC for this model was effectively zero, indicating that variation in choosing health facility delivery attributable to LGA clustering was negligible when geopolitical zone was accounted for. The LR test was omitted in this model due to lack of available estimate.

In line with the inclusion criteria of $p < 0.25$, the following variables qualify for inclusion in the multivariable multilevel logistic regression model (Models 2 and 4): maternal age group, formal education, level of education, marital status, ethnicity, husband's occupation, ownership of transportation, age at first birth, gravidity, parity, number of children alive, birth order, prior adverse birth outcomes, booking status, pregnancy-related complications, pregnancy outcome, residence and zone. These variables exhibited either statistically significant associations or clustering effects (ICC and LR test) suggesting relevance at the community level (LGA). The random intercept models confirm the appropriateness of multilevel modelling.

## Multilevel Logistic Regression Analysis of Factors Influencing Facility-Based Delivery

**Table 15: Multivariable multilevel logistic regression analysis for individual and community level factors influencing women's choice of place of delivery among reproductive age women 15-49 who deliver last child at any health facilities (n = 902)**

| | Model 1 Null model | | Model 2 Individual level | | Model 3 Community level | | Model 4 Full model | |
|---|---|---|---|---|---|---|---|---|
| **Predictors[a]** | **AOR (95% CI)** | **P-value** | **AOR (95% CI)** | **P-value** | **AOR (95% CI)** | **P-value** | **AOR (95% CI)** | **P-value** |
| *Constant* | 3.32 (2.70 - 4.09) | **<0.0001** | 2.62 (0.06 - 114.9) | 0.617 | 1.85 (0.90 - 3.81) | 0.092 | 2.09 (0.04 - 105.9) | 0.712 |
| *Individual level factors* | | | | | | | | |
| **Age:** | | | | | | | | |
| 15 – 19 years | | | Ref | - | | | Ref | - |
| 20 – 24 years | | | 1.90 (0.10 - 34.7) | 0.663 | | | 1.81 (0.09 - 33.57) | 0.691 |
| 25 – 29 years | | | 0.99 (0.06 - 16.4) | 0.996 | | | 0.94 (0.06 - 15.75) | 0.964 |
| 30 – 34 years | | | 1.89 (0.11 - 32.7) | 0.661 | | | 1.77 (0.10 - 31.09) | 0.698 |
| 35 – 39 years | | | 1.94 (0.11 - 34.8) | 0.654 | | | 1.75 (0.09 - 32.12) | 0.705 |
| 40 – 44 years | | | 1.50 (0.08 - 27.8) | 0.785 | | | 1.33 (0.07 - 25.33) | 0.846 |
| 45 – 49 years | | | 0.93 (0.05 - 17.7) | 0.959 | | | 0.83 (0.04 - 16.12) | 0.900 |
| **Formal education:** | | | | | | | | |
| No formal education | | | Ref | - | | | Ref | - |
| Had formal education | | | 1 | *na*[b] | | | 1 | *na*[b] |
| **Highest completed level of education:** | | | | | | | | |
| Primary | | | Ref | - | | | Ref | - |
| Secondary | | | 1.73 (0.78 - 3.81) | 0.173 | | | 1.71 (0.77 - 3.76) | 0.185 |
| Post-secondary | | | 4.06 (1.69 - 9.74) | **0.002** | | | 3.87 (1.61 - 9.32) | **0.003** |
| Post-tertiary | | | 12.5 (2.33 - 67.7) | **0.003** | | | 12.7 (2.37 - 68.16) | **0.003** |
| **Marital Status:** | | | | | | | | |
| Married | | | Ref | - | | | Ref | - |
| Single | | | 0.81 (0.12 - 5.70) | 0.833 | | | 0.79 (0.11 - 5.49) | 0.815 |
| Divorced | | | 1.08 (0.08 - 14.07) | 0.956 | | | 1.18 (0.09 - 15.60) | 0.900 |

| | | | | |
|---|---|---|---|---|
| **Ethnicity:** | | | | |
| Yoruba | Ref | - | Ref | - |
| Ibo | 1.54 (0.46 - 5.15) | 0.482 | 1.55 (0.45 - 5.31) | 0.488 |
| Hausa | 0.32 (0.03 - 4.00) | 0.377 | 0.26 (0.02 - 3.18) | 0.290 |
| Others | 0.28 (0.04 - 2.07) | 0.212 | 0.26 (0.04 - 1.89) | 0.184 |
| **Husband's occupation:** | | | | |
| Unemployed | Ref | - | Ref | - |
| Self-employed/Entrepr. | 1.34 (0.25 - 7.09) | 0.728 | 1.39 (0.27 - 7.37) | 0.692 |
| Private/Professional | 1.21 (0.21 - 6.84) | 0.833 | 1.28 (0.23 - 7.25) | 0.781 |
| Civil Servant | 1.58 (0.28 - 9.08) | 0.610 | 1.64 (0.28 - 9.39) | 0.581 |
| **Ownership of means of transportation:** | | | | |
| Yes | Ref | - | Ref | - |
| No | 0.79 (0.49 - 1.28) | 0.341 | 0.78 (0.48 - 1.28) | 0.326 |
| **Age at first birth:** | | | | |
| 15 – 19 years | Ref | - | Ref | - |
| 20 – 24 years | 0.69 (0.26 - 1.89) | 0.728 | 0.71 (0.27 - 1.89) | 0.492 |
| 25 – 29 years | 0.31 (0.11 - 0.88) | 0.833 | 0.31 (0.11 - 0.86) | **0.025** |
| 30 – 34 years | 2.19 (0.43 - 11.02) | 0.610 | 2.23 (0.45 - 11.17 | 0.327 |
| 35 – 39 years | 1 | *na*[b] | 1 | *na*[b] |
| **Gravidity:** | | | | |
| Primigravida | Ref | - | Ref | - |
| Multigravida | 0.58 (0.21 - 1.54) | 0.273 | 0.54 (0.19 - 1.47) | 0.226 |
| **Parity** | | | | |
| Primiparous | Ref | - | Ref | - |
| Multiparous | 0.59 (0.12 - 3.09) | 0.540 | 0.63 (0.12 - 3.29) | 0.586 |
| **Number of children alive:** | | | | |
| 0 | Ref | - | Ref | - |
| 1-4 | 0.62 (0.21 - 1.81) | 0.379 | 0.65 (0.22 - 1.92) | 0.435 |
| 5-8 | 1 | *na*[b] | 1 | *na*[b] |

| | | | | | | |
|---|---|---|---|---|---|---|
| **Birth Order of the Last Child:** | | | | | | |
| 0 | Ref | - | | | Ref | - |
| 1 | 3.13 (0.45 - 21.8) | 0.249 | | | 2.84 (0.40 - 20.07) | 0.295 |
| 2-3 | 5.67 (1.35 - 23.8) | **0.018** | | | 5.25 (1.23 - 22.34) | **0.025** |
| 4-5 | 3.79 (0.95 - 15.08 | 0.059 | | | 3.54 (0.88 - 14.27) | 0.075 |
| 6+ | 1 | na[b] | | | 1 | *na*[b] |
| **Adverse birth outcomes during delivery preceding the last pregnancy:** | | | | | | |
| Yes | Ref | - | | | Ref | - |
| No | 1.25 (0.59 - 2.65) | 0.568 | | | 1.20 (0.57 - 2.56) | 0.632 |
| **Booking status:** | | | | | | |
| Booked | Ref | - | | | Ref | - |
| Unbooked | 0.08 (0.04 - 0.15) | **<0.0001** | | | 0.08 (0.04 - 0.15) | **<0.0001** |
| **Pregnancy related problem during the last pregnancy:** | | | | | | |
| Yes | Ref | - | | | Ref | - |
| No | 0.65 (0.32 - 1.32) | 0.238 | | | 0.58 (0.28 - 1.21) | 0.149 |
| **Outcome of the last pregnancy:** | | | | | | |
| Live birth | Ref | - | | | Ref | - |
| Stilll-birth | 1 | *na*[b] | | | 1 | *na*[b] |
| Died few days after | 1 | *na*[b] | | | 1 | *na*[b] |
| *Community level fators* | | | | | | |
| **Residence:** | | | | | | |
| Urban | | | Ref | - | Ref | - |
| Rural | | | 1.33 (0.67 - 2.65) | 0.417 | 1.34 (0.59 - 3.00) | 0.480 |
| **Zone:** | | | | | | |
| Osun Central | | | Ref | - | Ref | - |
| Osun East | | | 2.36 (1.20 - 4.61) | **0.012** | 1.69 (0.73 -3.91) | 0.217 |

*a. This model estimates the odds of choosing health facility as the choice of place of delivery over other possible choices.*

b. *Omitted*

c. *P-value is significant at 0.05*

A two-level multilevel logistic regression model was fitted to determine individual and community-level factors associated with women's choice of place of delivery. The hierarchical structure of the data, with individual women nested within Local Government Areas (LGAs), necessitated the use of random intercept models to account for intra-cluster correlation. Four models were fitted sequentially: Model 1 (null model without predictors), Model 2 (adjusted for individual-level factors), Model 3 (adjusted for community-level factors), and Model 4 (the full model adjusted for both individual- and community-level covariates).

In the null model (Model 1), the odds of facility delivery were significantly different across LGAs, as shown by the intercept-only model with an odds ratio of 3.32 (95% CI: 2.70–4.09, $p < 0.0001$), indicating that clustering by LGA contributed to variation in delivery location. This supported the justification for using a multilevel modelling approach.

In Model 2, which adjusted for individual-level characteristics, none of the maternal age categories was significantly associated with facility delivery. Compared to women aged 15–19 years, those aged 20–24, 25–29, 30–34, 35–39, 40–44, and 45–49 years showed adjusted odds ratios (AORs) ranging from 0.93 to 1.94, but none reached statistical significance (all $p > 0.65$). Educational attainment, however, was strongly associated with facility delivery. Compared to women with no formal education, those with post-secondary and post-tertiary education were significantly more likely to deliver in a health facility. Specifically, women with secondary education had 4.06 times the odds (95% CI: 1.69–9.74, $p = 0.002$), and those with post-tertiary education had 12.5 times the odds (95% CI: 2.33–67.7, $p = 0.003$) of facility-based delivery.

Marital status and ethnicity were not significantly associated with place of delivery in the individual-level model. Neither were husband's occupation nor ownership of transportation. Age at first birth showed that women who had their first child between 20 and 24 years had significantly lower odds of facility delivery compared to those who gave birth at 15–19 years (AOR = 0.31, 95% CI: 0.11–0.88, p = 0.025). Other categories of age at first birth did not show significant associations.

Parity-related variables, including gravidity, parity, number of living children, and birth order, did not exhibit strong associations with facility delivery in the individual model. However, birth order of 2–3 showed borderline significance (AOR = 5.67, 95% CI: 1.35–23.8, p = 0.018) compared to the reference category of first birth. Adverse birth outcomes preceding the index pregnancy and pregnancy-related complications were not statistically significant.

A very strong and statistically significant predictor of facility delivery was booking status. Women who did not book for antenatal care had substantially reduced odds of delivering in a health facility (AOR = 0.08, 95% CI: 0.04–0.15, p < 0.0001). This finding remained consistent in the full model.

Model 3, which adjusted for community-level variables only, showed that women residing in Osun East were significantly more likely to deliver in a health facility compared to those in Osun Central (AOR = 2.36, 95% CI: 1.20–4.61, p = 0.012). However, place of residence (urban vs. rural) was not a significant predictor (AOR = 1.33, 95% CI: 0.67–2.65, p = 0.417).

In the final model (Model 4), which incorporated both individual and community-level predictors, the effect of education persisted. Women with secondary and post-tertiary education remained significantly more likely to utilise health facilities for

delivery. Post-tertiary education, in particular, retained its strong effect with an AOR of 12.7 (95% CI: 2.37–68.16, p = 0.003). Age at first birth between 20–24 years also remained significantly associated with lower odds of facility delivery (AOR = 0.31, 95% CI: 0.11–0.86, p = 0.025). Booking status remained a strong independent predictor, with unbooked women continuing to show dramatically lower odds of health facility delivery (AOR = 0.08, 95% CI: 0.04–0.15, p < 0.0001).

Community-level significance of geopolitical zone attenuated in the full model. Women from Osun East were still more likely to use facilities, but the association was no longer statistically significant (AOR = 1.69, 95% CI: 0.73–3.91, p = 0.217), suggesting confounding by individual-level characteristics.

Across the models, random intercept variance at the LGA level was minimal, indicating that the variability in the likelihood of facility delivery between LGAs was small after accounting for observed individual and community-level characteristics. This was confirmed by a low intraclass correlation (ICC), suggesting that most of the variation resided at the individual level.

Overall, multilevel modelling showed that women's education level, age at first birth, and booking status were the strongest independent determinants of facility-based delivery, while geopolitical zone contributed modestly when not adjusted for individual-level factors.

**Table 16: Model comparison for determinants of facility delivery among reproductive age women 15-49 who deliver last child at any health facilities (n = 902)**

| Comparator | Null (Model 1) | Model 2 | Model 3 | Model 4 |
|---|---|---|---|---|
| n | 902 | 692 | 902 | 692 |
| df | 2 | 31 | 3 | 33 |
| Community variance (LGA) | 0.086 | 1.95e-33 | 1.18e-32 | 1.13e-35 |
| 95% CI | (lower, upper) | | | |
| ICC (%) | 0.025 | 0.0 | 0.0 | 0.0 |
| PCV (%) | Reference | 100.0 | 100.0 | 100.0 |
| MOR | 1.323 | 1.00 | 1.00 | 1.00 |
| Log-likelihood | -489.0167 | -270.36 | -482.83 | -269.43 |
| AIC | 982.0335 | 602.733 | 971.6795 | 604.8661 |
| BIC | 991.6427 | 743.4602 | 986.0933 | 754.6724 |

Bivariate multilevel logistic regression analysis was conducted to examine the association between maternal age group and facility-based delivery, while accounting for clustering at the level of Local Government Areas (LGAs). The model was specified as a two-level random-intercept logistic regression, which allowed individual responses (Level 1) to be nested within LGAs (Level 2), thus accounting for potential intraclass correlation across communities. This approach ensures that unobserved heterogeneity attributable to the community (LGA) level is appropriately modeled, improving the accuracy of the estimated fixed effects.

Model comparison statistics were used to assess the relative fit of four nested models: the null model (Model 1) containing only the random intercept, the individual-level model (Model 2) including sociodemographic and reproductive factors, the community-level model (Model 3) including geographical variables, and the full

model (Model 4) combining both individual and community-level predictors. The results of the multilevel logistic regression model comparison for determinants of facility delivery among women of reproductive age (15–49 years) who delivered their last child in any health facility are summarised in Table 16. The null model (Model 1), which contained no explanatory variables, yielded a community-level variance (LGA-level) of 0.086, indicating that a small proportion of the variability in facility delivery was attributable to differences between LGAs. The corresponding intraclass correlation coefficient (ICC) was 2.5%, suggesting that only a minimal proportion of the variance in the likelihood of facility delivery was due to clustering at the LGA level in the absence of individual- or community-level predictors. The median odds ratio (MOR) for the null model was 1.323, indicating modest heterogeneity between LGAs.

When individual-level factors were introduced in Model 2, the community variance dropped sharply to $1.95 \times 10^{-33}$, effectively eliminating between-LGA variability. This corresponds to an ICC of 0%, suggesting that individual-level characteristics fully explained the variance observed in the null model. The proportional change in variance (PCV) relative to the null model was 100%, further confirming that all LGA-level variation was accounted for once individual-level factors were included. The MOR also reduced to 1.00, indicating no residual clustering effect. The model fit improved substantially, with the log-likelihood increasing from –489.02 in Model 1 to –270.36 in Model 2.

Model 3, which included only community-level predictors, also resulted in the complete elimination of LGA-level variance ($1.18 \times 10^{-32}$), yielding an ICC of 0% and a PCV of 100%. As with Model 2, the MOR reduced to 1.00, suggesting that the included community-level factors alone explained the clustering effect present in the

null model. The log-likelihood improved from the null model to –482.83 in Model 3, although the gain in model fit was less pronounced than that observed with individual-level factors.

In the final model (Model 4), which incorporated both individual- and community-level predictors, the community-level variance further approached zero ($1.13 \times 10^{-35}$), maintaining an ICC of 0% and a PCV of 100%. The MOR remained 1.00, again indicating the absence of meaningful LGA-level heterogeneity. Model 4 achieved the highest log-likelihood (–269.43), suggesting the best overall model fit among the four.

Based on the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC), Model 2 had the lowest AIC (602.73) and BIC (743.46), indicating the best fit among all models. Although Model 4 incorporated more variables and had a marginally better log-likelihood than Model 2 (−269.43 vs. −270.37), the increase in parameter count led to a higher AIC (604.87) and BIC (754.67), suggesting possible overfitting. Model 3, which included only community-level variables, did not substantially improve fit compared to the null model, with AIC and BIC values of 971.68 and 986.09, respectively.

In summary, these findings indicate that the clustering of facility delivery outcomes at the LGA level observed in the null model was completely explained when either individual-level, community-level, or both sets of predictors were included. The final model's complete elimination of between-cluster variance and superior log-likelihood values underscore the dominant explanatory power of the included predictors.

Furthermore, the bivariate multilevel analysis revealed no significant age-related differences in the odds of facility delivery, and model comparison results suggested that individual-level factors had greater explanatory power than community-level

variables. These findings support the inclusion of individual characteristics in multivariate models, while also justifying the use of multilevel modelling due to the statistically significant LGA-level clustering.

**Reported RMC (RMC) Received**

**Table 17: Types of Reported RMC Received During Childbirth at health facilities among women age 15-49 (n = 619)**

| RMC Domains | Strongly disagree n (%) | Disagree n (%) | Neutral n (%) | Agree n (%) | Strongly agree n (%) |
|---|---|---|---|---|---|
| **A. Friendly care** | | | | | |
| 1. Healthcare workers approached me in a kind manner (very nice and ready to help) | 14 (2.3) | 11 (1.8) | 28 (4.5) | 310 (50.1) | 256 (41.4) |
| 2. Healthcare workers were friendly (pleasant) in their treatment | 13 (2.1) | 8 (1.3) | 33 (5.3) | 326 (52.7) | 239 (38.6) |
| 3. Healthcare workers sounded positively/ encouraging on topics about pain and its relief | 14 (2.3) | 12 (1.9) | 55 (8.9) | 338 (54.6) | 200 (32.3) |
| 4. Healthcare workers were concerned and empathic. | 12 (1.9) | 17 (2.8) | 41 (6.6) | 333 (53.8) | 216 34.9) |
| 5. Healthcare workers respected me as an individual | 10 (1.6) | 17 (2.8) | 33 (5.3) | 351 (56.7) | 208 (33.6) |
| 6. Healthcare workers communicated to me in a language within my understanding | 10 (1.6) | 13 (2.1) | 27 (4.4) | 323 (52.2) | 246 (39.7 |
| 7. Healthcare workers addressed me by my name | 79 (12.8) | 96 (15.5) | 101 (16.3) | 277 (44.8) | 66 (10.7) |
| **B. Abuse-free care** | | | | | |
| 1. Healthcare workers responded to my needs in a professionally expected manner | 15 (2.4) | 17 (2.8) | 40 (6.5) | 374 (60.4) | 173 (28.0) |
| 2. Healthcare workers hit me during delivery | 303 (49.0) | 221 (35.7) | 33 (5.3) | 46 (7.4) | 16 (2.6) |
| 3. Healthcare workers yelled at me because I did not do what they wanted me to do | 262 (42.3) | 206 (33.3) | 61 (9.9) | 63 (10.2) | 27 (4.4) |
| **C. Timely care** | | | | | |
| 1. I was not given prompt service and I waited for a long time. | 239 (38.6) | 255 (41.2) | 60 (9.7) | 49 (7.9) | 16 (2.6) |
| 2. I was free to practice my cultural and religious traditions in the facility | 125 (20.2) | 41 (6.6) | 115 (18.6) | 296 (47.8) | 42 (6.8) |
| 3. Problems in the health facility often delay their attending to their patients | 215 (34.7) | 226 (36.5) | 83 (13.4) | 83 (13.4) | 12 (1.9) |
| **D. Non-discriminatory care** | | | | | |
| 1. Some healthcare workers mistreated me because of some of my personal characteristics. (for example, my age, my financial status). | 195 (31.5) | 279 (45.1) | 52 (8.4) | 39 (6.3) | 54 (8.7) |
| **2.** Some healthcare workers offended me and my companions due to my personal characteristics (for example, my age, my financial status). | 202 (32.6) | 273 (44.1) | 51 (8.2) | 38 (6.1) | 55 (8.9) |

## Friendly Care

Overall, respondents reported highly positive experiences related to friendly and respectful care. A total of 310 women (50.1%) agreed and 256 women (41.4%) strongly agreed that healthcare workers approached them in a kind and helpful manner, yielding a high overall mean score of 4.3 (±0.82). Similarly, 52.7% agreed and 38.6% strongly agreed that providers were pleasant in their treatment, with a mean of 4.2 (±0.79).

Positive communication was also noted. Over half (54.6%) agreed and 32.3% strongly agreed that staff were encouraging when discussing pain relief, giving a mean score of 4.1 (±0.82). A comparable proportion, 53.8% agreed and 34.9% strongly agreed, said the staff were concerned and empathetic (mean = 4.17 ± 0.82). Respect for individuality was affirmed by 56.7% agreeing and 33.6% strongly agreeing, while only 1.6% strongly disagreed (mean = 4.18 ± 0.78).

Language was generally not a barrier: 52.2% agreed and 39.7% strongly agreed that they were communicated with in a language they understood, resulting in a mean score of 4.26 (±0.78).

However, a less positive result emerged regarding being addressed by name. While 44.8% agreed and 10.7% strongly agreed, a significant number of respondents—12.8% strongly disagreed and 15.5% disagreed—said they were not called by name, reflecting a lower mean score of 3.3 (±1.22).

**Abuse-Free Care**

The majority of women experienced respectful and non-abusive treatment. A total of 60.4% agreed and 28.0% strongly agreed that healthcare workers responded to their needs in a professionally expected manner, yielding a high mean of 4.1 (±0.82).

However, negative behaviours were reported by a minority. 49.0% strongly disagreed and 35.7% disagreed that they were hit during delivery, but 2.6% strongly agreed, and 7.4% agreed, raising concern. The mean score for this item was relatively low at 1.8 (±1.01).

Similarly, yelling was denied by 42.3% (strongly disagree) and 33.3% (disagree), though 10.2% agreed and 4.4% strongly agreed they were yelled at during delivery. This yielded a mean score of 2.0 (±1.15).

**Timely Care**

This domain revealed significant dissatisfaction regarding timeliness of care. A combined 79.8% (strongly disagree/disagree) indicated they did not have to wait long, suggesting that waiting was not a widespread issue; however, only 10.5% (agree/strongly agree) reported prompt service, resulting in a low mean score of 1.9 (±1.02).

Regarding the freedom to practise cultural or religious traditions, responses were more varied. While 47.8% agreed, 20.2% strongly disagreed, and 18.6% were neutral, resulting in a moderate mean of 3.1 (±1.27).

When asked if problems in the facility caused delays, 71.2% (strongly disagree/disagree) said they did not experience such delays. However, 13.4% agreed, and 13.4% were neutral, giving a mean of 2.1 (±1.09).

**Non-Discriminatory Care**

Most women did not experience discrimination, although a minority reported concerning behaviours. Regarding mistreatment due to personal characteristics (e.g.,

age or financial status), 31.5% strongly disagreed, and 45.1% disagreed, while 6.3% agreed and 8.7% strongly agreed. The mean score was relatively low at 2.2 (±1.19).

Similarly, for offensive behaviour toward the respondent or her companions, 32.6% strongly disagreed, 44.1% disagreed, but 8.9% strongly agreed, again suggesting that while the majority were treated respectfully, discrimination was not absent. The mean score was 2.1 (±1.20).

Overall, the highest scoring RMC domain was friendly care, particularly around kind communication and respectful interaction. However, addressing women by name and timely care had noticeably lower satisfaction levels. Reports of physical or verbal abuse were low, though not negligible, and some women experienced or perceived discrimination, particularly based on personal characteristics.

**Table 18: Types of Reported RMC Received During Childbirth at health facilities among women age 15-49 (n = 619)**

| RMC Domains | Mean±SD | Percentage of the maximum score Mean±SD |
|---|---|---|
| **A. Friendly care [7 items]** | **4.1±0.6** | **76.8±14.8** |
| 1. Healthcare workers approached me in a kind manner (very nice and ready to help) | 4.3±0.82 | 81.6±20.4 |
| 2. Healthcare workers were friendly (pleasant) in their treatment | 4.2±0.79 | 81.1±19.7 |
| 3. Healthcare workers sounded positively/ encouraging on topics about pain and its relief | 4.1±0.82 | 78.2±20.6 |
| 4. Healthcare workers were concerned and empathic. | 4.17±0.82 | 19.2±20.5 |
| 5. Healthcare workers respected me as an individual | 4.18±0.78 | 79±.5±19.6 |
| 6. Healthcare workers communicated to me in a language within my understanding | 4.26±0.78 | 81.6±19.4 |
| 7. Healthcare workers addressed me by my name | 3.3±1.22 | 56.3±30.4 |
| **B. Abuse-free care [3 items]** | **2.6±0.6** | **40.7±16.0** |
| 1. Healthcare workers responded to my needs in a professionally expected manner | 4.1±0.82 | 77.2±20.4 |
| 2. Healthcare workers hit me during delivery | 1.8±1.01 | 19.7±25.4 |
| 3. Healthcare workers yelled at me because I did not do what they wanted me to do | 2.0±1.15 | 25.24±28.8 |
| **C. Timely care [3 items]** | **2.4±0.9** | **35.0±21.5** |
| 1. I was not given prompt service and I waited for a long time. | 1.9±1.02 | 23.7±25.4 |
| 2. I was free to practice my cultural and religious traditions in the facility | 3.1±1.27 | 53.6±31.7 |
| 3. Problems in the health facility often delay their attending to their patients | 2.1±1.09 | 27.8±27.2 |
| **D. Non-discriminatory care [2 items]** | **2.2±1.2** | **28.8±29.3** |
| 1. Some healthcare workers mistreated me because of some of my personal characteristics. (for example, my age, my financial status). | 2.2±1.19 | 28.9±29.7 |
| 2. Some healthcare workers offended me and my companions due to my personal characteristics (for example, my age, my financial status). | 2.1±1.20 | 28.6±29.9 |
| **Overall Reported RMC received** | **4.8±5.2** | **54.8±8.6** |

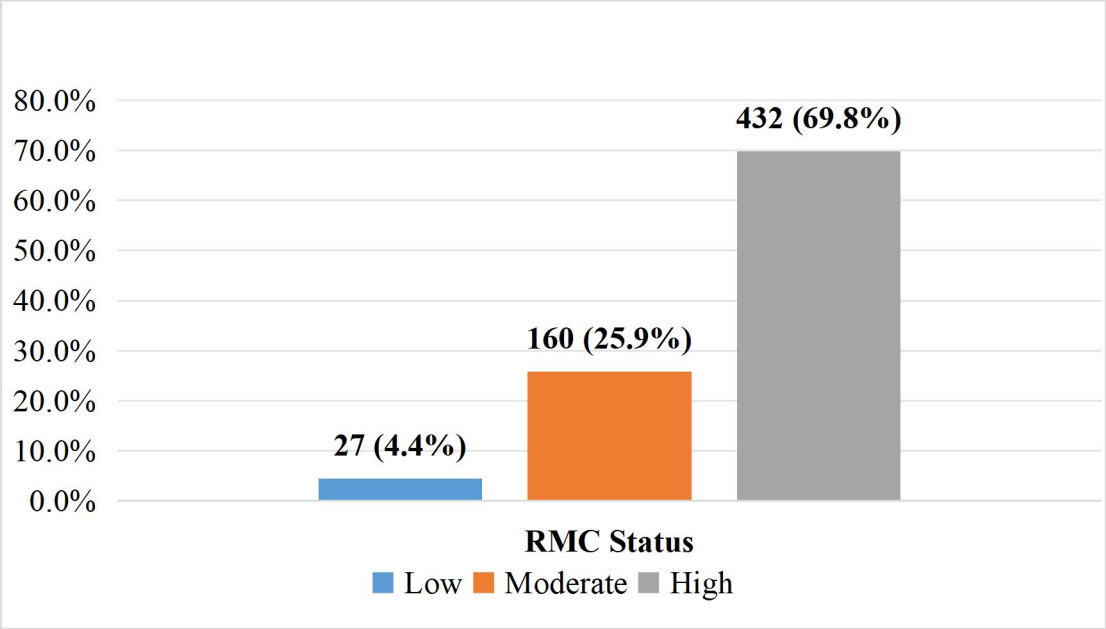


**Figure 4.1: Levels of reported friendly RMC domain received at Health Facilities**

The analysis of the *friendly maternity care* subcomponent revealed that a majority of the women reported high levels of respectful care during childbirth. Out of 619 respondents, 432 women, representing 69.8% of the sample, were categorised as having received a high level of friendly care, defined as scores equal to or greater than 75%. A further 160 women, accounting for 25.9%, reported a moderate level of friendly care with scores ranging from 50% to 74%. Only 27 women, which constitutes 4.4% of the total, experienced low levels of friendly care, scoring below 50%. These findings suggest that while a substantial proportion of women reported receiving friendly maternity care, there remains a smaller group for whom the experience was less satisfactory.

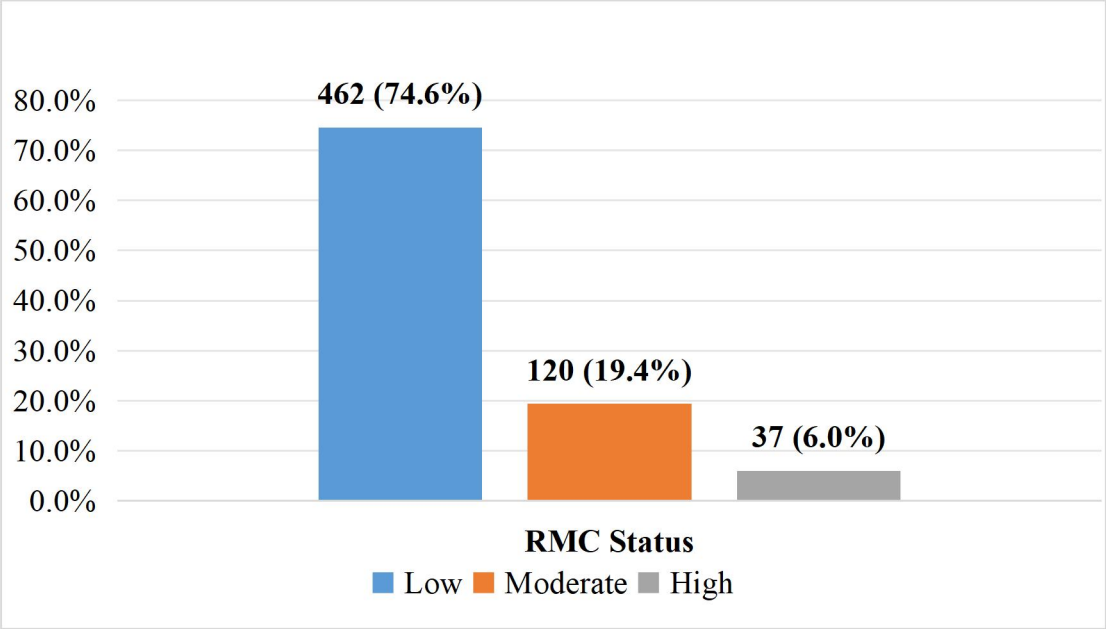


**Figure 4.2: Levels of reported abuse-free RMC domain received at Health Facilities**

The findings for the *abuse-free maternity care* subcomponent indicate that a substantial proportion of women reported experiencing low levels of abuse-free care during childbirth. Out of the 619 women surveyed, 462, representing 74.6% of the sample, were classified as having received low levels of abuse-free care, with scores below 50%. An additional 120 women, accounting for 19.4%, reported moderate levels of abuse-free care, with scores ranging from 50% to 74%. Only 37 women, which makes up 6.0% of the total sample, reported high levels of abuse-free care, with scores of 75% or higher. These results highlight significant gaps in the provision of maternity care that is free from mistreatment or abuse.

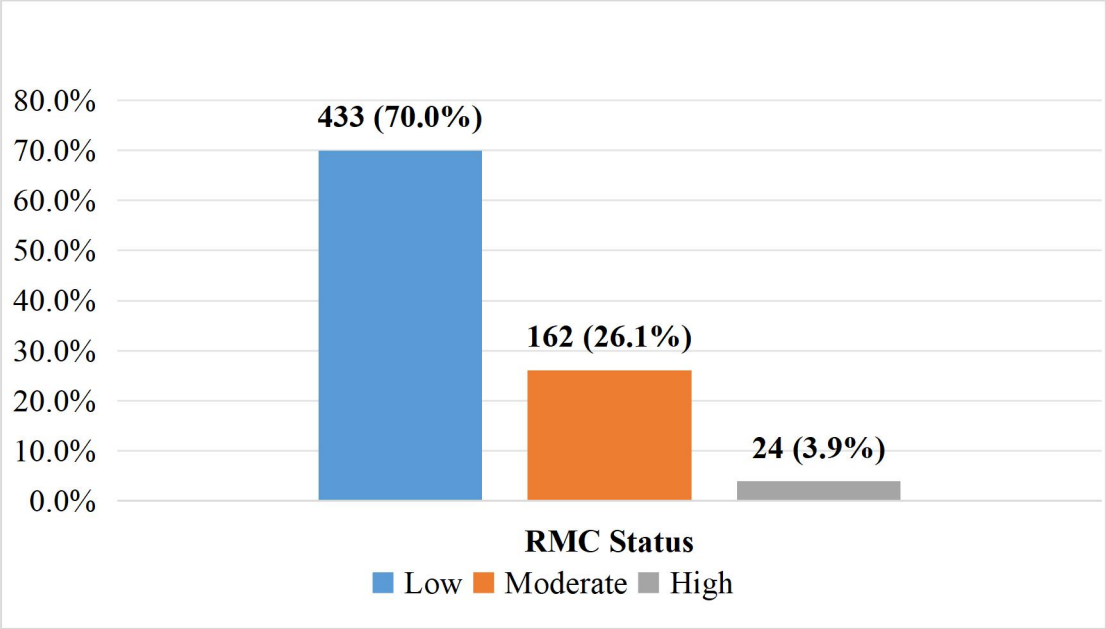


**Figure 4.3: Levels of reported timely-care RMC domain received at Health Facilities**

The analysis of the *timely care* subcomponent of RMC shows that the majority of respondents reported low levels of timely care during childbirth. Among the 619 women included in the study, 433, representing 70.0% of the sample, were classified as having received low levels of timely care, defined by scores below 50%. A further 162 women, accounting for 26.2%, reported moderate levels of timely care, with scores between 50% and 74%. Only 24 women, comprising 3.9% of the sample, experienced high levels of timely care, with scores equal to or above 75%. These findings suggest that delays or a lack of promptness in care provision remain a major concern in the childbirth experience of many women.

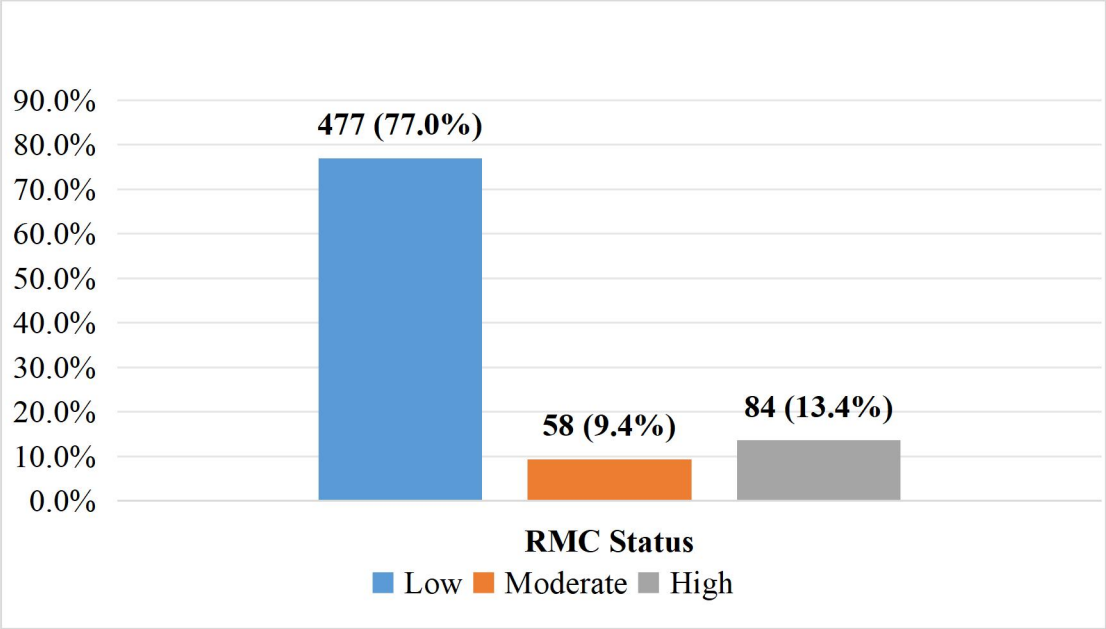


**Figure 4.4: Levels of reported non-discriminatory care RMC domain received at Health Facilities**

The assessment of the *non-discriminatory care* subcomponent reveals that a considerable majority of women reported low levels of respectful care in this domain. Of the 619 respondents, 477 women, representing 77.1% of the sample, were categorised as having received low levels of non-discriminatory care, with scores falling below 50%. An additional 58 women, accounting for 9.4%, reported moderate levels of care, with scores between 50% and 74%. Only 84 women, making up 13.6% of the total, reported high levels of non-discriminatory care, defined by scores of 75% or higher. These results indicate that discriminatory practices remain a significant challenge in the maternity care setting for a large proportion of the women surveyed.

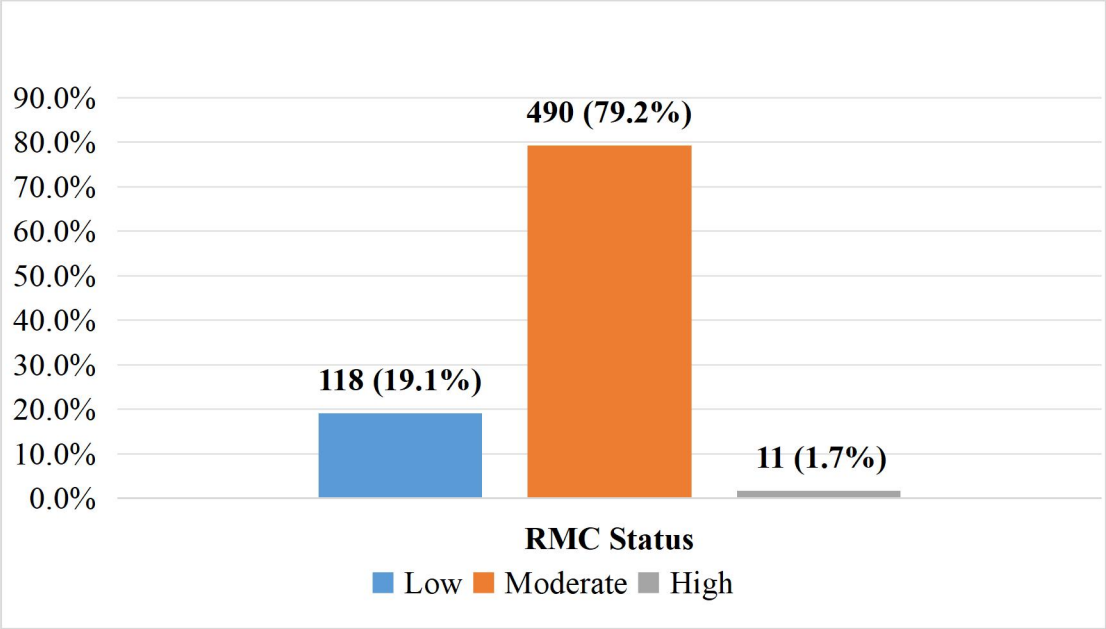


**Figure 4.5: Levels of reported overall RMC domain received at Health Facilities**

The overall analysis of RMC (RMC) indicates that the vast majority of women experienced care that did not reach the highest standards of respectful treatment during childbirth. Out of 619 respondents, only 11 women, representing 1.8% of the sample, were classified as having received high levels of RMC, with percentage scores equal to or above 75%. A significantly larger proportion, 490 women or 79.2%, reported moderate levels of respectful care, with scores ranging from 50% to 74%. Meanwhile, 118 women, accounting for 19.1% of the sample, received low levels of RMC, with scores below 50%. These findings highlight a critical need for improvements in maternity care practices to ensure more women consistently receive care that upholds dignity, respect, and quality.

## Factors influencing reported RMC status among women in the region

**Table 17: Percent distribution of reproductive age women 15-49 by who deliver last child at any health facilities and their choice of place of delivery of last child according to research area (n = 619)**

| Research area | Level of Reported RMC Received | | | Number of women |
|---|---|---|---|---|
| | Low RMC (<50%) | Moderate RMC (50% - <70%) | High RMC (≥75%) | |
| **Residence:** | | | | |
| Urban | 14.1 | 85.1 | 0.8 | 262 |
| Rural | 22.7 | 74.8 | 2.5 | 357 |
| *P-value* | | ***0.0005*** | | |
| **Zone:** | | | | |
| Osun Central | 23.5 | 73.5 | 3.0 | 302 |
| Osun East | 14.8 | 84.5 | 0.6 | 317 |
| *P-value* | | ***0.001*** | | |
| **LGA:** | | | | |
| Boluwaduro | 22.6 | 77.4 | 0.0 | 31 |
| Ifelodun | 21.9 | 78.1 | 0.0 | 32 |
| Odo Otin | 31.4 | 62.9 | 5.7 | 35 |
| Boripe | 20.8 | 75.0 | 4.2 | 24 |
| Ila | 26.9 | 65.4 | 7.7 | 26 |
| Ifedayo | 23.5 | 73.5 | 2.9 | 34 |
| Irepodun | 13.8 | 82.8 | 3.5 | 29 |
| Olorunda | 10.7 | 89.3 | 0.0 | 28 |
| Osogbo | 12.2 | 87.8 | 0.0 | 41 |
| Atakunmosa East | 23.5 | 73.5 | 2.9 | 34 |
| Atakunmosa West | 22.6 | 77.4 | 0.0 | 31 |
| Ife East | 18.9 | 81.1 | 0.0 | 37 |
| Ife North | 32.1 | 64.3 | 3.6 | 28 |
| Ife South | 6.1 | 90.9 | 3.0 | 33 |
| Ife Central | 21.4 | 78.6 | 0.0 | 42 |
| Obokun | 13.3 | 86.7 | 0.0 | 30 |
| Oriade | 17.4 | 82.6 | 0.0 | 23 |
| Ilesa East | 18.8 | 79.2 | 2.1 | 48 |
| Ilesa West | 6.1 | 93.9 | 0.0 | 33 |
| *P-value* | | *0.357* | | |
| **Total** | **118 (19.1)** | **490 (79.2)** | **11 (1.8)** | **619 (100.0)** |

Among the 619 women surveyed, 19.1% (n = 118) reported receiving a low level of Respectful Maternity Care (RMC), defined as receiving below 50% of RMC indicators. A majority of respondents, 79.2% (n = 490), experienced a moderate level of RMC (50% to <75%), while only 1.8% (n = 11) received a high level of RMC (≥75%).

With respect to residence, there was a statistically significant association between place of residence and the reported level of RMC received ($p = 0.0005$). In urban areas, a substantial majority of women (85.1%) received moderate RMC, with only 14.1% reporting low RMC, and a minimal proportion (0.8%) receiving high RMC. In contrast, women in rural areas were more likely to report low RMC at 22.7%, while 74.8% received moderate RMC, and 2.5% received high RMC. The data suggest that rural residence may be associated with reduced quality of maternity care in terms of respectful treatment.

Analysis by geopolitical zone also revealed a significant relationship between zone and RMC experience ($p = 0.001$). In Osun Central, 23.5% of women reported low RMC, compared to 14.8% in Osun East. Conversely, a greater proportion of women in Osun East reported moderate RMC (84.5%) than those in Osun Central (73.5%). Notably, a slightly higher percentage of respondents in Osun Central (3.0%) received high RMC compared to Osun East (0.6%). These findings point to zonal disparities in the experience of maternity care, possibly reflecting systemic or facility-level differences.

At the level of local government areas (LGAs), variation in reported RMC was more nuanced, though the association between LGA and level of RMC was not statistically significant ($p = 0.357$). The proportion of women who received low RMC ranged widely across LGAs, from as high as 32.1% in Ife North to as low as 6.1% in Ife South and Ilesa West. The highest proportion of moderate RMC was recorded in Ilesa West (93.9%), Obokun (90.9%), and Olorunda (89.3%). High levels of RMC were scarcely reported across LGAs, with percentages not exceeding 7.7% in any area. In some LGAs such as Osogbo, Atakunmosa East, and Ife East, no respondents reported

high RMC at all. These results underscore the need for targeted interventions that address intra-state inequalities in maternity care experiences.

Overall, the analysis highlights that while most women reported receiving moderate levels of RMC, a considerable fraction—particularly those in rural areas and Osun Central—experienced low RMC, and very few women overall received high RMC. This suggests both geographic and systemic gaps in the implementation of respectful maternity care standards across Osun State.

**Table 19: Percent distribution of reproductive age women 15-49 by who deliver last child at any health facilities and their choice of place of delivery of last child according to women's background characteristics (n = 619)**

| Background characteristics | Level of Reported RMC Received | | | Number of women |
|---|---|---|---|---|
| | Low RMC (<50%) | Moderate RMC (50% - <70%) | High RMC (≥75%) | |
| **Age:** | | | | |
| 15 – 19 years | 25.0 | 75.0 | 0.0 | 4 |
| 20 – 24 years | 17.0 | 80.9 | 2.1 | 47 |
| 25 – 29 years | 20.2 | 79.8 | 0.0 | 104 |
| 30 – 34 years | 19.6 | 76.9 | 3.5 | 143 |
| 35 – 39 years | 20.0 | 79.3 | 0.7 | 150 |
| 40 – 44 years | 14.3 | 83.9 | 1.8 | 112 |
| 45 – 49 years | 23.7 | 72.9 | 3.4 | 59 |
| *P-value* | | *0.675* | | |
| **Formal education:** | | | | |
| No formal education | 30.0 | 70.0 | 0.0 | 90 |
| Had formal education | 17.2 | 80.7 | 2.1 | 529 |
| *P-value* | | ***0.008*** | | |
| **Marital Status:** | | | | |
| Married | 19.4 | 78.9 | 1.7 | 594 |
| Single | 15.8 | 84.2 | 0.0 | 19 |
| Divorced | 0.0 | 83.3 | 16.7 | 5 |
| *P-value* | | *0.056* | | |
| **Ethnicity:** | | | | |
| Yoruba | 18.7 | 79.7 | 1.6 | 572 |
| Ibo | 22.6 | 71.0 | 6.4 | 31 |
| Hausa | 25.0 | 75.0 | 0.0 | 8 |
| Others | 25.0 | 75.0 | 0.0 | 8 |
| | | *0.533* | | |
| **Husband's occupation:** | | | | |
| Unemployed | 17.4 | 73.9 | 8.7 | 23 |
| Self-employed/Entrepr. | 20.0 | 78.7 | 1.3 | 385 |
| Private/Professional | 17.9 | 79.0 | 3.1 | 95 |
| Civil Servant | 17.2 | 81.9 | 0.9 | 116 |
| *P-value* | | *0.117* | | |
| **Ownership of means of transportation:** | | | | |
| Yes | 19.8 | 78.4 | 1.9 | 430 |
| No | 17.5 | 81.0 | 1.5 | 189 |
| *P-value* | | *0.766* | | |
| **Total** | **118 (19.1)** | **490 (79.2)** | **11 (1.8)** | **619 (100.0)** |

Across the total sample, 19.1% (n = 118) of women reported low RMC (<50%), 79.2% (n = 490) reported moderate RMC (50% to <75%), and only 1.8% (n = 11) reported receiving high RMC (≥75%).

In terms of age, there was no statistically significant association between age group and reported RMC ($p = 0.675$). Among adolescents aged 15–19 years, one in four (25.0%) experienced low RMC, while 75.0% received moderate RMC and none reported high RMC. A relatively lower proportion of low RMC was reported among women aged 40–44 years (14.3%), with 83.9% receiving moderate care and 1.8% reporting high RMC. The highest percentage of high RMC (3.5%) was reported among women aged 30–34 years, while women aged 25–29 years had no reported incidence of high RMC. These results suggest no consistent pattern of RMC levels across age categories.

The level of formal education showed a statistically significant association with the quality of RMC received ($p = 0.008$). Among women with no formal education, 30.0% reported low RMC, compared with 17.2% among those who had formal education. Additionally, 2.1% of women with formal education received high RMC, whereas none of the uneducated respondents reported high RMC. This suggests that women with formal education were more likely to report receiving moderate to high levels of respectful maternity care compared to their uneducated counterparts.

There was no statistically significant relationship between marital status and RMC received, although the association approached significance ($p = 0.056$). Among married women, 19.4% reported low RMC, 78.9% moderate RMC, and 1.7% high RMC. Among single women, 15.8% reported low RMC, 84.2% moderate, and none reported high RMC. Interestingly, among the few women who were divorced ($n = 5$), none reported low RMC, 83.3% received moderate care, and 16.7% received high RMC. The elevated percentage of high RMC among divorced women, though based on a small number, may warrant further qualitative investigation.

Ethnic background was not significantly associated with the level of RMC received (p = 0.533). Among Yoruba women—who comprised the majority of the sample—18.7% reported low RMC and 1.6% high RMC. Comparatively, Ibo women were more likely to report low RMC at 22.6% and also had a slightly higher rate of high RMC at 6.4%. Hausa women reported 25.0% low RMC and none reported high RMC. Similarly, women from other ethnic groups also reported 25.0% low RMC with no high RMC observed.

Regarding husband's occupation, no statistically significant association was observed with RMC levels (p = 0.117). However, the data reveal slight differences across categories. Among women whose husbands were unemployed, 17.4% reported low RMC and 8.7% reported high RMC—the highest proportion of high RMC in this category. Women whose husbands were civil servants had the lowest proportion of low RMC at 17.2%, with 81.9% reporting moderate care and less than 1% reporting high RMC. High RMC was also reported by 3.1% of those whose husbands worked in the private or professional sector.

The ownership of a means of transportation showed no significant association with the level of RMC received (p = 0.766). Among women who had access to transport, 19.8% reported low RMC, 78.4% moderate, and 1.9% high RMC. Those without access to transportation reported slightly lower low RMC (17.5%), with 81.0% receiving moderate RMC and 1.5% high RMC. The findings suggest that transport access does not substantially influence the perceived respectfulness of care received during delivery.

In summary, the level of formal education was the only background characteristic significantly associated with the quality of RMC received, highlighting the protective

role of education in maternal care experiences. Other factors including age, marital status, ethnicity, husband's occupation, and transport ownership did not show statistically significant associations, although trends in the data suggest areas for further exploration, particularly the higher report of high RMC among divorced women and those whose husbands were unemployed.

**Table 20: Percent distribution of reproductive age women 15-49 by who deliver last child at any health facilities and their choice of place of delivery of last child according to women's obstetrics profile (n = 619)**

| Obstetrics profile | Level of Reported RMC Received | | | Number of women |
|---|---|---|---|---|
| | Low RMC (<50%) | Moderate RMC (50% - <70%) | High RMC (≥75%) | |
| **Age at first birth:** | | | | |
| 15 – 19 years | 22.5 | 77.6 | 0.0 | 49 |
| 20 – 24 years | 18.5 | 79.5 | 2.0 | 254 |
| 25 – 29 years | 18.0 | 80.3 | 1.7 | 234 |
| 30 – 34 years | 20.3 | 70.0 | 2.7 | 74 |
| 35 – 39 years | 37.5 | 62.5 | 0.0 | 8 |
| *P-value* | | *0.876* | | |
| **Gravidity:** | | | | |
| Primigravida | 22.2 | 77.0 | 0.7 | 135 |
| Multigravida | 18.2 | 79.8 | 2.1 | 484 |
| | | *0.360* | | |
| **Parity** | | | | |
| Primiparous | 22.9 | 76.5 | 0.6 | 166 |
| Multiparous | 17.7 | 80.1 | 2.2 | 453 |
| | | *0.157* | | |
| **Number of children alive:** | | | | |
| 0 | 0.0 | 100.0 | 0.0 | 2 |
| 1-4 | 19.4 | 78.9 | 1.7 | 578 |
| 5-8 | 15.4 | 82.1 | 2.6 | 39 |
| | | *0.906* | | |
| **Birth Order of the Last Child:** | | | | |
| 0 | 33.3 | 66.7 | 0.0 | 3 |
| 1 | 22.2 | 77.2 | 0.6 | 171 |
| 2-3 | 18.5 | 79.0 | 2.5 | 324 |
| 4-5 | 17.3 | 80.8 | 1.9 | 104 |
| 6+ | 5.9 | 94.1 | 0.0 | 17 |
| | | *0.619* | | |
| **Adverse birth outcomes during delivery preceding the last pregnancy:** | | | | |
| Yes | 22.8 | 72.8 | 4.4 | 92 |
| No | 18.4 | 80.3 | 1.3 | 527 |
| | | *0.068* | | |
| **Booking status:** | | | | |
| Booked | 18.6 | 79.7 | 1.7 | 582 |
| Unbooked | 27.0 | 70.3 | 2.7 | 37 |
| | | *0.388* | | |
| **Pregnancy related problem during the last pregnancy:** | | | | |
| Yes | 31.4 | 68.6 | 0.0 | 102 |
| No | 16.6 | 81.2 | 2.1 | 517 |
| | | *0.001* | | |
| **Outcomer of the last pregnancy:** | | | | |
| Live birth | 18.7 | 79.5 | 1.8 | 609 |
| Stilll-birth | 42.9 | 57.1 | 0.0 | 7 |
| Died few days after | 33.3 | 66.7 | 0.0 | 3 |
| *P-value* | | *0.539* | | |
| **Total** | **118 (19.0)** | **490 (79.2)** | **11 (1.8)** | **619 (100.0)** |

Among the total sample, 19.0% (n = 118) of women reported receiving low levels of RMC (<50%), while 79.2% (n = 490) received moderate RMC (50%–<70%). Only a small proportion (1.8%, n = 11) reported high RMC (≥75%).

An analysis of age at first birth showed no statistically significant association with the level of RMC received (p = 0.876). Among women who had their first child between ages 15–19 years, 22.5% reported low RMC and none experienced high RMC. In contrast, women whose first birth occurred between ages 25–29 reported a slightly lower prevalence of low RMC (18.0%) and 1.7% received high RMC. The highest level of high RMC was observed among women who gave birth for the first time at age 30–34 (2.7%), despite having a slightly higher rate of low RMC at 20.3%. Although there is no consistent trend across age groups, there appears to be a general clustering of moderate RMC across all categories.

With respect to gravidity, 22.2% of primigravidae (first pregnancy) reported low RMC, compared to 18.2% of multigravidae. Conversely, high RMC was more frequently reported among multigravidae (2.1%) than primigravidae (0.7%). The association between gravidity and RMC, however, did not reach statistical significance (p = 0.360).

A similar pattern was observed with parity, where 22.9% of primiparous women (first birth) experienced low RMC, compared to 17.7% among multiparous women. Only 0.6% of primiparous women reported high RMC, whereas this was slightly higher (2.2%) among multiparous women. The relationship between parity and RMC level was not statistically significant (p = 0.157), but the data suggest that women with previous childbirth experience may encounter more respectful care during subsequent deliveries.

Considering the number of living children, there was no significant relationship with reported RMC ($p = 0.906$). Although low RMC was slightly more common among women with one to four children (19.4%) compared to those with five to eight children (15.4%), the variations were marginal. None of the women with no living children ($n = 2$) reported low RMC.

The birth order of the last child also did not show a statistically significant association with RMC levels ($p = 0.619$). Women whose last child was their first birth reported low RMC at 22.2%, while those with higher-order births (6 or more) had the lowest prevalence of low RMC at 5.9%. High RMC was most common among women with birth order 2–3 (2.5%) and 4–5 (1.9%), while those with birth orders 0, 1, and 6+ had no reports of high RMC.

Adverse outcomes in the pregnancy preceding the last delivery (such as stillbirth or early neonatal death) were not significantly associated with the level of RMC received ($p = 0.068$), though the pattern is worth noting. Women who had experienced such outcomes were more likely to report low RMC (22.8%) compared to those who did not (18.4%). High RMC was slightly more prevalent among women without a history of adverse outcomes (1.3%) versus those who had experienced one (4.4%).

Analysis of booking status—whether the woman was booked for antenatal care at the facility where she delivered—showed no statistically significant difference in RMC levels ($p = 0.388$). Among booked women, 18.6% reported low RMC, 79.7% received moderate RMC, and 1.7% experienced high RMC. In comparison, among unbooked women, the prevalence of low RMC was higher at 27.0%, and the rate of high RMC was also slightly higher at 2.7%. Despite this, the statistical test does not confirm a strong association.

A significant relationship was found between experience of pregnancy-related problems during the last pregnancy and the level of RMC received ($p = 0.001$). Among women who reported such problems, nearly one-third (31.4%) experienced low RMC, compared to 16.6% of women who did not report such complications. Notably, none of the women with pregnancy-related problems reported high RMC, while 2.1% of women without such problems did. These results suggest that women experiencing complications are more likely to perceive their care as disrespectful or suboptimal.

The outcome of the last pregnancy (live birth, stillbirth, or neonatal death) did not have a statistically significant association with RMC levels ($p = 0.539$). Among women whose pregnancy resulted in live birth, 18.7% reported low RMC and 1.8% reported high RMC. In contrast, 42.9% of those who experienced stillbirth and 33.3% of those whose baby died shortly after birth reported low RMC, and none in these groups reported high RMC. Although the sample sizes for stillbirths and neonatal deaths were small ($n = 7$ and $n = 3$, respectively), these findings reinforce the possible link between perceived disrespectful care and poor maternal or neonatal outcomes.

In summary, the only obstetric factor significantly associated with the level of Respectful Maternity Care was the experience of pregnancy-related problems during the last pregnancy. Women who experienced complications were significantly more likely to report low RMC and less likely to report high RMC, pointing to a critical area of concern for healthcare quality. Other obstetric indicators such as gravidity, parity, birth order, and booking status did not show statistically significant associations, though observable trends suggest that prior reproductive experience and facility engagement may have some influence on RMC outcomes.

## To determine the factors influencing reported RMC scores during health facility childbirth in the region

**Table 21: Linear regression model showing location, women characteristics and obstetrics profiles predictors of reported RMC scores among age women 15-49 who deliver last child at any health facilities (n = 619)**

| Predictors[a] | Coeff. | P-value[c] | 95% CI |
|---|---|---|---|
| **Residence:** | | | |
| Urban | Ref | - | - |
| Rural | -8.056 | 0.225 | -21.08 - 4.97 |
| **Zone:** | | | |
| Osun Central | Ref | - | - |
| Osun East | -9.819 | 0.120 | -22.19 - 2.56 |
| **LGA:** | | | |
| Boluwaduro | Ref | - | - |
| Ifelodun | 1.723 | 0.423 | -2.50 - 5.95 |
| Odo Otin | -1.122 | 0.594 | -5.25 - 3.00 |
| Boripe | 0.444 | 0.846 | -4.06 - 4.94 |
| Ila | 2.357 | 0.301 | -2.11 - 6.82 |
| Ifedayo | 0.766 | 0.716 | -3.36 - 4.89 |
| Irepodun | 2.336 | 0.290 | -1.99 - 6.67 |
| Olorunda | 1.956 | 0.371 | -2.33 - 6.24 |
| Osogbo | 1.192 | 0.547 | -2.69 - 5.07 |
| Atakunmosa East | 2.002 | 0.346 | -2.16 - 6.17 |
| Atakunmosa West | -1.550 | 0.471 | -5.76 - 2.66 |
| Ife East | -0.491 | 0.810 | -4.50 - 3.52 |
| Ife North | 0.038 | 0.986 | -4.37 - 4.45 |
| Ife South | 3.405 | 0.103 | -0.68 - 7.49 |
| Ife Central | -0.610 | 0.758 | -4.50 - 3.28 |
| Obokun | 7.973 | 0.232 | -5.11 - 21.06 |
| Oriade | 7.478 | 0.265 | -5.69 - 20.64 |
| Ilesa East | 1.471 | 0.449 | -2.34 - 5.28 |
| Ilesa West | 0[b] | - | - |
| **Age:** | | | |
| 15 – 19 years | Ref | - | - |
| 20 – 24 years | -6.465 | 0.164 | -15.57 - 2.64 |
| 25 – 29 years | -9.034 | **0.049** | -18.01 - -0.05 |
| 30 – 34 years | -7.950 | 0.085 | -16.99 - 1.09 |
| 35 – 39 years | -7.565 | 0.104 | -16.69 - 1.57 |
| 40 – 44 years | -6.877 | 0.145 | -16.12 - 2.36 |
| 45 – 49 years | -7.874 | 0.102 | -17.32 - 1.57 |
| **Formal education:** | | | |
| No formal education | Ref | - | - |
| Had formal education | 3.894 | **<0.0001** | 1.75 - 6.03 |
| **Marital Status:** | | | |
| Married | Ref | - | - |
| Single | 3.538 | 0.192 | -1.78 - 8.86 |
| Divorced | 5.790 | 0.142 | -1.94 - 13.52 |
| **Ethnicity:** | | | |
| Yoruba | Ref | - | - |
| Ibo | 2.826 | 0.076 | -0.29 - 5.95 |
| Hausa | 0.324 | 0.921 | -6.12 - 6.77 |
| Others | -0.911 | 0.792 | -7.70 - 5.88 |
| **Husband's occupation:** | | | |
| Unemployed | Ref | - | - |
| Self-employed/Entrepr. | -1.785 | 0.489 | -6.85 - 3.28 |

| | | | |
|---|---|---|---|
| Private/Professional | -0.683 | 0.799 | -5.94 - 4.58 |
| Civil Servant | -1.504 | 0.575 | -6.76 - 3.76 |
| **Ownership of means of transportation:** | | | |
| Yes | Ref | - | - |
| No | -0.306 | 0.702 | -1.88 - 1.26 |
| **Age at first birth:** | | | |
| 15 – 19 years | Ref | - | - |
| 20 – 24 years | 1.502 | 0.285 | -1.25 - 4.26 |
| 25 – 29 years | 1.674 | 0.256 | -1.21 - 4.56 |
| 30 – 34 years | 2.099 | 0.241 | -1.41 - 5.61 |
| 35 – 39 years | -3.391 | 0.312 | -9.97 - 3.19 |
| **Gravidity:** | | | |
| Primigravida | Ref | - | - |
| Multigravida | 3.426 | **0.043** | 0.11 - 6.73 |
| **Parity** | | | |
| Primiparous | Ref | - | - |
| Multiparous | -0.741 | 0.759 | -5.47 - 3.99 |
| **Number of children alive:** | | | |
| 0 | Ref | - | - |
| 1-4 | -14.439 | 0.054 | -29.15 - 0.27 |
| 5-8 | -12.852 | 0.098 | -28.06 - 2.36 |
| **Birth Order of the Last Child:** | | | |
| 0 | Ref | - | - |
| 1 | -6.875 | 0.358 | -21.54 - 7.79 |
| 2-3 | -7.376 | 0.333 | -22.31 - 7.56 |
| 4-5 | -8.254 | 0.281 | -23.29 - 6.78 |
| 6+ | -7.003 | 0.376 | -22.54 - 8.52 |
| **Adverse birth outcomes during delivery preceding the last pregnancy:** | | | |
| Yes | Ref | - | - |
| No | -1.873 | 0.076 | -3.94 - 0.19 |
| **Booking status:** | | | |
| Booked | Ref | - | - |
| Unbooked | 0.944 | 0.523 | -1.96 - 3.85 |
| **Pregnancy related problem during the last pregnancy:** | | | |
| Yes | Ref | - | - |
| No | 6.37 | **<0.0001** | 4.38 - 8.34 |
| **Outcomer of the last pregnancy:** | | | |
| Live birth | Ref | - | - |
| Stilll-birth | -5.029 | 0.147 | -11.83 - 1.77 |
| Died few days after | -8.947 | 0.194 | -2.46 - 4.56 |
| **Constant** | 83.107 | **<0.0001** | 59.55 - 106.65 |

*a. Dependet variable = RMC Percent scores*

b. *Omitted*

c. *P-value is significant at 0.05*

This regression model identified formal education, gravidity, age group 20–24 years, and pregnancy-related complications as statistically significant predictors of Respectful Maternity Care scores. Education and gravidity were positively associated with higher RMC, while younger maternal age (20–24 years) was associated with a

lower score. Interestingly, women who experienced complications during pregnancy reported significantly higher RMC scores, potentially indicating more targeted attention by health providers. Other variables, including residence, ethnicity, parity, transport access, and marital status, showed no significant independent associations in this model.

In the adjusted linear regression model, the constant RMC score was 83.11 ($p < 0.0001$; 95% CI: 59.55 to 106.65), representing the predicted RMC score for the reference group across all covariates.

In terms of residence, women living in rural areas had an RMC score that was on average 8.06 points lower than their urban counterparts, although this difference was not statistically significant ($p = 0.225$; 95% CI: –21.08 to 4.97).

Across geopolitical zones, women residing in Osun East had RMC scores that were 9.82 points lower than those in Osun Central, but this association was not statistically significant ($p = 0.120$; 95% CI: –22.19 to 2.56).

With respect to local government areas (LGAs), none of the LGAs showed statistically significant differences in RMC scores compared to the reference LGA, Boluwaduro. Coefficients for the LGAs ranged from –1.55 (Atakunmosa East) to +7.97 (Ife Central), all with p-values above 0.05, indicating no strong evidence of LGA-level variation in RMC score.

Regarding age, compared to the youngest group (15–19 years), women aged 20–24 years had a significantly lower RMC score by 9.03 points ($p = 0.049$; 95% CI: –18.01 to –0.05). Other age categories also had lower RMC scores ranging from –6.47 to –7.95, though these were not statistically significant.

Formal education emerged as a significant predictor of RMC score. Women who had formal education scored 3.89 points higher on the RMC scale than those without any formal education ($p < 0.0001$; 95% CI: 1.75 to 6.03).

For marital status, single women had a higher RMC score compared to married women by 3.54 points, and divorced women by 5.79 points, although these differences were not statistically significant ($p = 0.192$ and $p = 0.142$, respectively).

In terms of ethnicity, Ibo women had RMC scores 2.83 points higher than Yoruba women ($p = 0.076$; 95% CI: –0.29 to 5.95), nearing statistical significance. Hausa and other ethnic groups showed no significant differences.

Husband's occupation was not significantly associated with RMC scores. Compared to women whose husbands were unemployed, the RMC scores were slightly lower among those whose husbands were self-employed (–1.79), in the private sector (–0.68), or civil servants (–1.50), with p-values all above 0.48.

Ownership of means of transportation was also not statistically associated with RMC scores. Women without transport scored 0.31 points lower than those with access ($p = 0.702$; 95% CI: –1.88 to 1.26).

Regarding age at first birth, none of the categories showed statistically significant differences in RMC scores compared to women who first gave birth at age 15–19. Women who gave birth at 30–34 years had the lowest coefficient (–3.39), while those at 25–29 had the highest (+2.10), all with p-values > 0.24.

Gravidity was significantly associated with RMC scores. Multigravidae women had RMC scores 3.43 points higher than primigravidae ($p = 0.043$; 95% CI: 0.11 to 6.73).

On the other hand, parity showed no significant association. Multiparous women had a slightly lower RMC score (–0.74; $p = 0.759$).

For number of living children, women with one to four children scored 14.44 points lower, and those with five to eight children scored 12.85 points lower than those with no living children, although these results narrowly missed statistical significance ($p = 0.054$ and $p = 0.098$, respectively).

When considering birth order of the last child, no category showed significant influence on RMC scores. Women whose last child was of order 6+ had slightly better scores than those with first or second births, but all coefficients had wide confidence intervals and non-significant p-values.

Regarding previous adverse birth outcomes, women who had no such outcomes had an RMC score 1.87 points lower than those who did, a difference that approached statistical significance ($p = 0.076$; 95% CI: –3.94 to 0.19).

Booking status was not associated with RMC scores. Unbooked women had a non-significant 0.94 point increase in score compared to booked women ($p = 0.523$).

The presence of pregnancy-related complications during the last pregnancy was significantly associated with higher RMC scores. Women who had complications scored on average 6.37 points higher than those without complications ($p < 0.0001$; 95% CI: 4.38 to 8.34), contrary to expectations and suggesting possible increased attention or care in complex cases.

Finally, regarding the outcome of the last pregnancy, women who had a stillbirth or lost the child shortly after birth had lower RMC scores (–5.03 and –8.95, respectively), although these were not statistically significant ($p = 0.147$ and $p = 0.194$, respectively).

**Table 22: Binary logistics regression model showing location, women characteristics and obstetrics profiles predicting the odds of receiving a better RMC [Moderate/High] vs Low reported RMC among reproductive age women 15-49 who deliver last child at any health facilities (n = 619)**

| | Bivariate Analysis | | | Multivariate Analysis | | |
|---|---|---|---|---|---|---|
| **Predictors[a]** | **Crude OR** | **P-value[c]** | **95% CI** | **Adjusted OR** | **P-value[c]** | **95% CI** |
| **Residence:** | | | | | | |
| Urban | Ref | - | - | Ref | - | - |
| Rural | 0.560 | **0.008** | 0.36 - 0.86 | 0.655 | 0.091 | 0.40 - 1.07 |
| **Age:** | | | | | | |
| 15 – 19 years | Ref | - | - | Ref | - | - |
| 20 – 24 years | 1.625 | 0.690 | 0.14 - 17.6 | 1.014 | 0.991 | 0.07 - 13.6 |
| 25 – 29 years | 1.317 | 0.815 | 0.13 - 13.3 | 0.617 | 0.708 | 0.05 - 7.71 |
| 30 – 34 years | 1.369 | 0.789 | 0.13 - 13.6 | 0.597 | 0.693 | 0.05 - 7.69 |
| 35 – 39 years | 1.333 | 0.806 | 0.13 - 13.2 | 0.557 | 0.659 | 0.04 - 7.42 |
| 40 – 44 years | 2.0 | 0.559 | 0.19 - 20.4 | 0.836 | 0.895 | 0.06 - 11.7 |
| 45 – 49 years | 1.071 | 0.954 | 0.10 - 11.1 | 0.365 | 0.461 | 0.03 - 5.32 |
| **Formal education:** | | | | | | |
| No formal education | Ref | - | - | Ref | - | - |
| Had formal education | 2.063 | **0.005** | 1.24 - 3.42 | 2.533 | **0.003** | 1.38 - 4.63 |
| **Marital Status:** | | | | | | |
| Married | Ref | - | - | Ref | - | - |
| Single | 1.281 | 0.698 | 0.36 - 4.46 | 2.339 | 0.356 | 0.38 - 14.2 |
| Divorced | 1 | *na*[b] | *na*[b] | 1 | *na*[b] | *na*[b] |
| **Ethnicity:** | | | | | | |
| Yoruba | Ref | - | - | Ref | - | - |
| Ibo | 0.789 | 0.592 | 0.33 - 1.87 | 0.703 | 0.463 | 0.27 - 1.79 |
| Hausa | 0.690 | 0.653 | 0.14 - 3.47 | 0.907 | 0.918 | 0.14 - 5.69 |
| Others | 0.690 | 0.653 | 0.14 - 3.47 | 0.792 | 0.844 | 0.08 - 8.03 |
| **Husband's occupation:** | | | | | | |
| Unemployed | Ref | - | - | Ref | - | - |
| Self-employed/Entrepr. | 0.842 | 0.761 | 0.27 - 2.55 | 2.118 | 0.370 | 0.41 - 10.9 |
| Private/Professional | 0.966 | 0.955 | 0.29 - 3.24 | 2.015 | 0.417 | 0.37 - 10.9 |
| Civil Servant | 1.011 | 0.986 | 0.31 - 3.29 | 1.968 | 0.434 | 0.36 - 10.7 |
| **Ownership of means of transportation:** | Ref | - | - | Ref | - | - |
| Yes | 1.165 | 0.501 | 0.75 - 1.82 | 1.129 | 0.638 | 0.68 - 1.87 |
| No | | | | | | |
| **Age at first birth:** | | | | | | |
| 15 – 19 years | Ref | - | - | Ref | - | - |
| 20 – 24 years | 1.275 | 0.521 | 0.61 - 2.68 | 1.479 | 0.357 | 0.64 - 3.40 |
| 25 – 29 years | 1.323 | 0.464 | 0.23 - 2.80 | 1.596 | 0.292 | 0.67 - 3.81 |
| 30 – 34 years | 1.139 | 0.772 | 0.47 - 2.74 | 1.239 | 0.695 | 0.42 - 3.63 |
| 35 – 39 years | 0.482 | 0.366 | 0.09 - 2.34 | 0.517 | 0.461 | 0.09 - 2.98 |
| **Gravidity:** | | | | | | |
| Primigravida | Ref | - | - | Ref | - | - |
| Multigravida | 1.285 | 0.291 | 0.81 - 2.05 | 0.824 | 0.697 | 0.31 - 2.18 |
| **Parity** | | | | | | |
| Primiparous | Ref | - | - | Ref | - | - |
| Multiparous | 1.384 | 0.143 | 0.89 - 2.14 | 2.353 | 0.238 | 0.57 - 9.74 |
| **Number of children alive:** | | | | | | |
| 0 | Ref | - | - | Ref | - | - |
| 1-4 | 0.756 | 0.541 | 0.31 - 1.85 | 1.037 | 0.953 | 0.32 - 3.38 |
| 5-8 | 1 | *na*[b] | *na*[b] | 1 | *na*[b] | *na*[b] |

| | | | | | | |
|---|---|---|---|---|---|---|
| **Birth Order of the Last Child:** | | | | | | |
| 0 | Ref | - | - | Ref | - | - |
| 1 | 1.750 | 0.651 | 0.15 - 19.8 | 1.822 | 0.791 | 0.02 - 154.7 |
| 2-3 | 2.200 | 0.523 | 0.19 - 24.7 | 1.706 | 0.816 | 0.02 - 152.5 |
| 4-5 | 2.380 | 0.487 | 0.21 - 27.8 | 1.989 | 0.766 | 0.02 - 185.4 |
| 6+ | 8.000 | 0.194 | 0.35 - 184.3 | 8.596 | 0.397 | 0.06 - 1248.6 |
| **Adverse birth outcomes during delivery preceding the last pregnancy:** | Ref | - | - | Ref | - | - |
| Yes | 1.311 | 0.320 | 0.76 - 2.24 | 0.879 | 0.689 | 0.46 - 1.65 |
| No | | | | | | |
| **Booking status:** | | | | | | |
| Booked | Ref | - | - | Ref | - | - |
| Unbooked | 0.615 | 0.207 | 0.29 - 1.31 | 0.554 | 0.164 | 0.24 - 1.27 |
| **Pregnancy related problem during the last pregnancy:** | | | | | | |
| Yes | Ref | - | - | Ref | - | - |
| No | 2.29 | **0.001** | 1.42 - 3.69 | 2.667 | **0.001** | 1.52 - 4.67 |
| **Outcomer of the last pregnancy:** | | | | | | |
| Live birth | Ref | - | - | Ref | - | - |
| Stilll-birth | 0.307 | 0.126 | 0.07 - 1.39 | 0.352 | 0.244 | 0.06 - 2.04 |
| Died few days after | 0.461 | 0.528 | 0.04 - 5.12 | 0.221 | 0.430 | 0.01 - 9.44 |
| **Constant** | | | | 0.246 | 0.613 | 0.001 - 55.69 |

*d. This model estimates the odds of receiving a better RMC [Moderate/High]=1, vs Low RMC=0.*
e. *Omitted*
f. *P-value is significant at 0.05*

Below is a detailed analysis report of Table 15: Binary logistic regression model predicting the likelihood of receiving better Respectful Maternity Care (RMC)—defined here as either moderate or high RMC (coded as 1)—as opposed to low RMC (coded as 0), among women aged 15–49 years who delivered their last child in a health facility in Osun State, Nigeria (n = 619).

In the bivariate analysis, women residing in rural areas had significantly lower odds of receiving better (moderate/high) RMC compared to their urban counterparts (crude OR = 0.560; p = 0.008; 95% CI: 0.36–0.86). However, after adjusting for covariates in the multivariate model, this association was no longer statistically significant, although the direction of the effect persisted (adjusted OR = 0.655; p = 0.091; 95% CI: 0.40–1.07). This suggests that location may be partly confounded by other sociodemographic or obstetric factors.

Age group was not a significant predictor of better RMC in either bivariate or multivariate models. Although women aged 40–44 had slightly higher odds in the crude model (OR = 2.0; p = 0.559), none of the age categories showed significant associations when compared to the reference group (15–19 years). In the multivariate analysis, the adjusted odds ratios ranged from 0.365 to 1.014, with all p-values > 0.29 and wide confidence intervals.

Formal education consistently emerged as a statistically significant and strong predictor of better RMC. Women who had formal education were over twice as likely to report better (moderate/high) RMC compared to those without formal education (crude OR = 2.063; p = 0.005; 95% CI: 1.24–3.42). This relationship remained significant after adjustment (adjusted OR = 2.533; p = 0.003; 95% CI: 1.38–4.63), underscoring the role of education in promoting respectful care.

Marital status was not a significant predictor in either analysis. Although single and divorced women had higher odds of receiving better RMC in the multivariate model (adjusted OR for single = 2.339; p = 0.356), these results were not statistically significant and were based on small subgroup sizes, limiting the reliability of these estimates.

Across ethnic groups, no statistically significant associations were found in either bivariate or multivariate analyses. The odds ratios for Ibo, Hausa, and other ethnic groups, compared to Yoruba women, remained close to 1.0 and non-significant, with confidence intervals crossing unity.

Husband's occupation did not show significant effects in either model. While adjusted odds ratios for women whose husbands were self-employed, professionals, or civil

servants were slightly above 2.0, none of these associations reached statistical significance (p-values between 0.370 and 0.434).

Ownership of a means of transportation was not significantly associated with better RMC. In the multivariate model, women without transport had slightly higher odds (adjusted OR = 1.129; p = 0.638), but the effect was not significant.

Age at first birth did not significantly predict better RMC in the multivariate model. Although women who had their first child at ages 20–34 had slightly higher odds (adjusted ORs ranging from 1.24 to 1.60), none of the associations were statistically significant (all p-values > 0.29).

Gravidity was not associated with better RMC. Women with multiple pregnancies had lower odds than primigravidae (adjusted OR = 0.824; p = 0.697). Similarly, parity showed no significant effect, although the adjusted odds for multiparous women were higher (OR = 2.353; p = 0.238), with wide confidence intervals indicating instability of the estimate.

The number of children alive did not predict better RMC. Women with 1–4 children had slightly higher odds compared to those with none (adjusted OR = 1.037; p = 0.953), but the result was non-significant.

No statistically significant associations were observed for birth order of the last child. Women with birth orders of 6 or more had the highest odds (adjusted OR = 8.596; p = 0.397), though the wide confidence interval (0.06–1248.6) reflects high uncertainty due to small sample size.

Women with no adverse birth outcomes in the pregnancy prior to the last were slightly less likely to report better RMC (adjusted OR = 0.879; p = 0.689), though the result was not statistically significant.

Booking status at delivery was not a significant predictor. Unbooked women had lower odds of receiving better RMC than booked women (adjusted OR = 0.554; p = 0.164; 95% CI: 0.24–1.27), though not significant.

A pregnancy-related complication during the last pregnancy was a strong and significant predictor of better RMC. In the bivariate model, women who reported such complications had more than twice the odds of receiving better RMC (crude OR = 2.29; p = 0.001; 95% CI: 1.42–3.69). This association remained robust after adjustment (adjusted OR = 2.667; p = 0.001; 95% CI: 1.52–4.67), suggesting increased provider attention or concern during complicated pregnancies.

The outcome of the last pregnancy (live birth vs. stillbirth or early neonatal death) did not significantly predict better RMC. Women who experienced stillbirth had an adjusted OR of 0.352 (p = 0.244), and those whose babies died shortly after had an OR of 0.221 (p = 0.430), though neither was statistically significant.

In summary, formal education and pregnancy-related complications during the last pregnancy were the only statistically significant and independent predictors of better Respectful Maternity Care after adjustment. Residence in rural areas was associated with lower odds of better RMC in unadjusted models, but this association diminished when controlling for other variables. Most sociodemographic and obstetric characteristics did not independently influence the likelihood of receiving better RMC, highlighting that education and clinical vulnerability (i.e. complications) may be the strongest drivers of provider responsiveness and respectful care in facility-based childbirth in Osun State.